Hyperspherical Contrastive Representations Unlock Planning and Exploration

Jonghae Park*, Sungil Son*, Jonghun Shin, Junseo Lee

H. Jin Kim, Benjamin Eysenbach

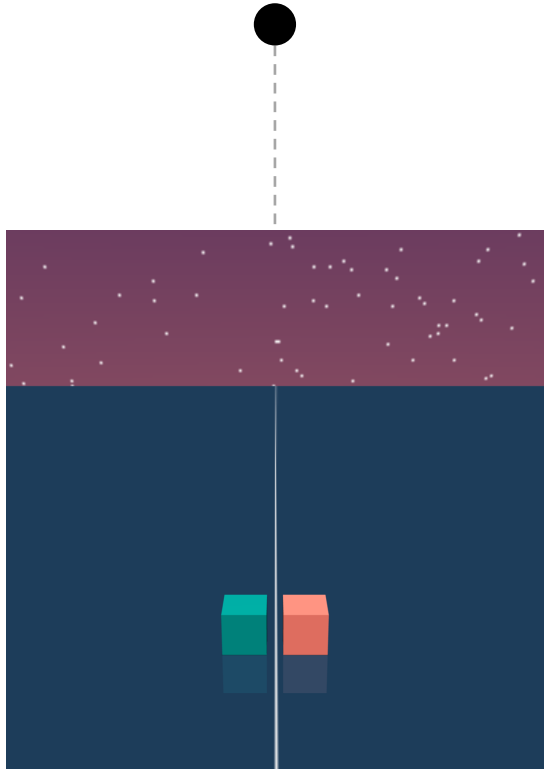


Consider a robot 🤖 asked to build a tower of blocks.

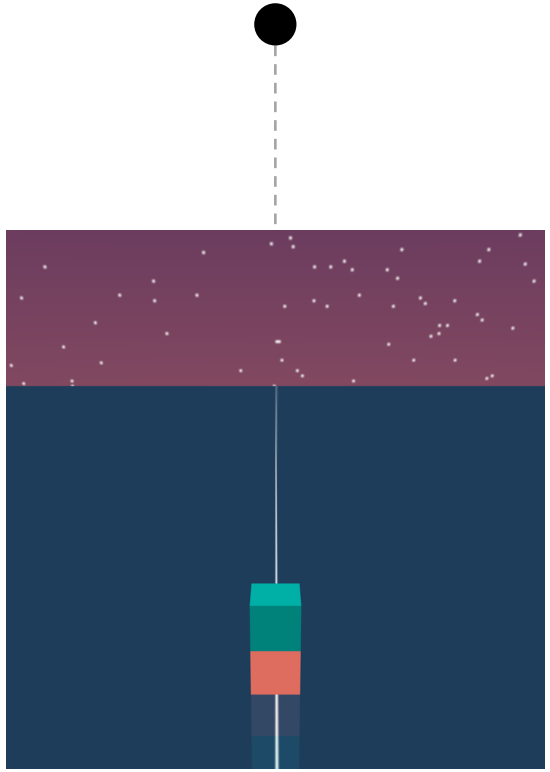
Consider a robot 🤖 asked to build a tower of blocks.

How many subgoals, *or waypoints,* does the robot need?

Stacking Two Blocks

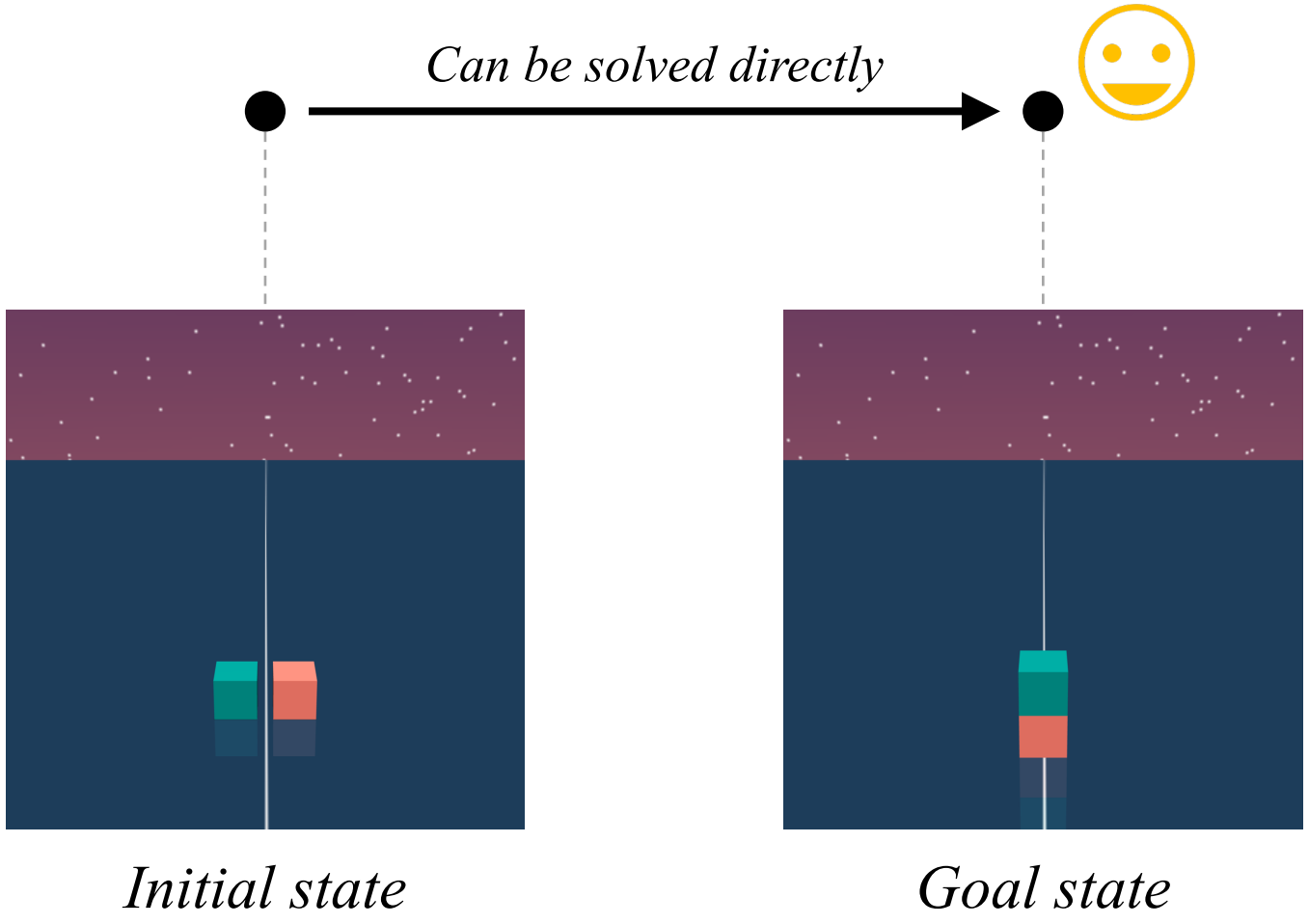


Initial state

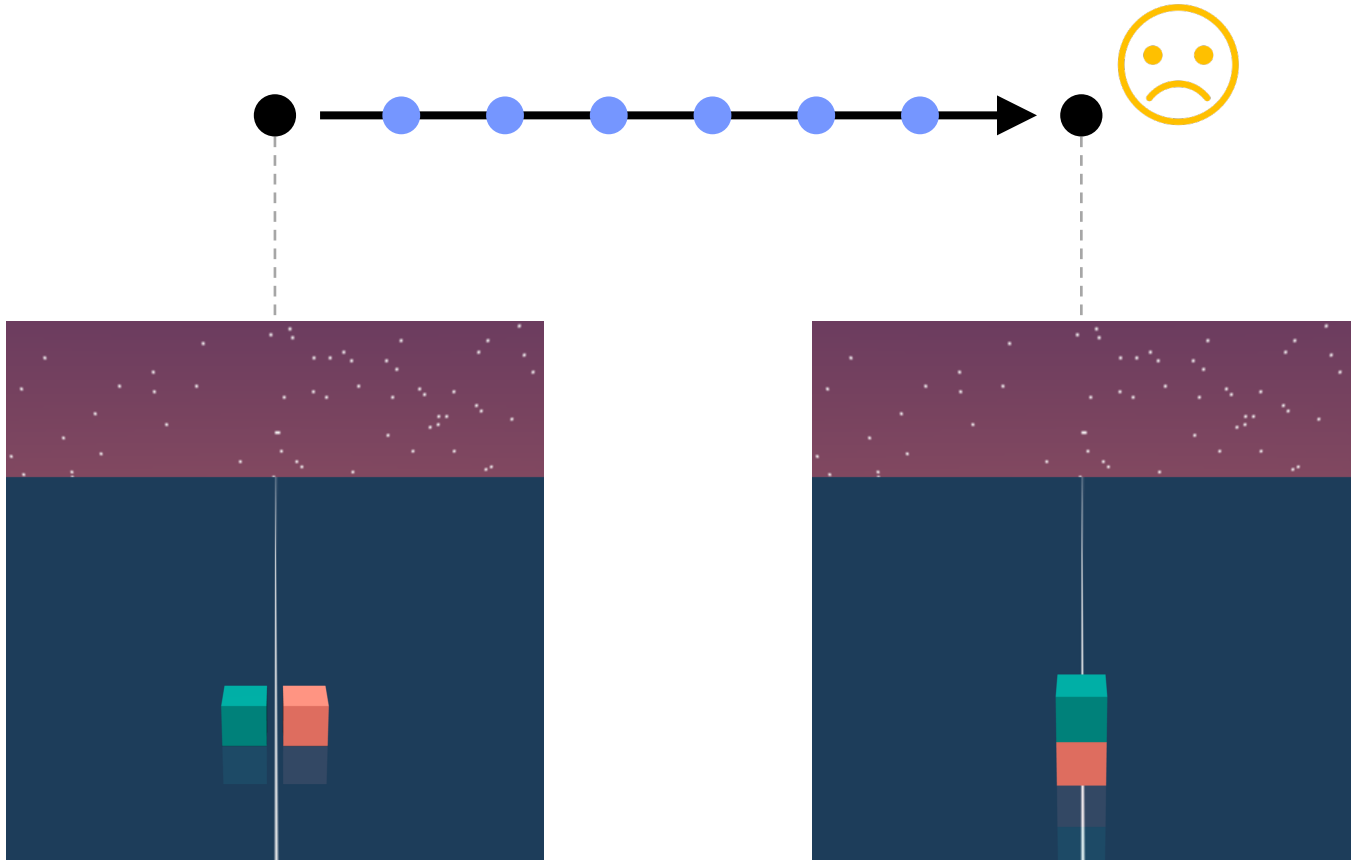


Goal state

Stacking Two Blocks

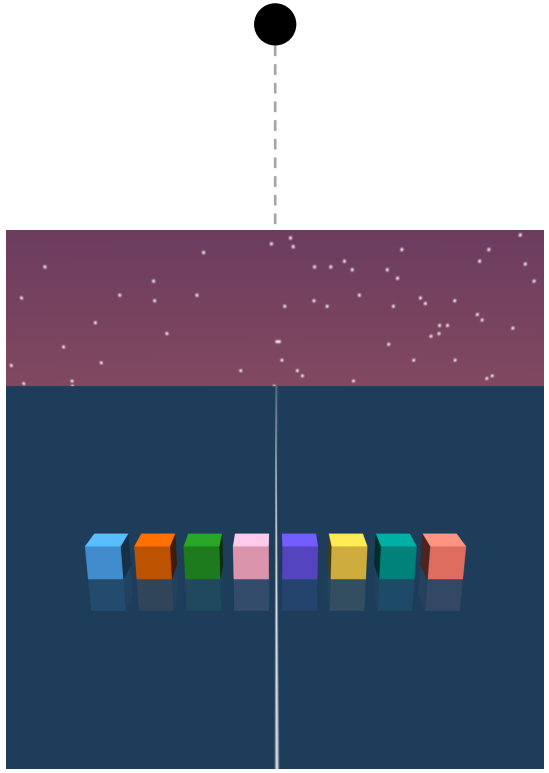


Stacking Two Blocks with too many planning stages

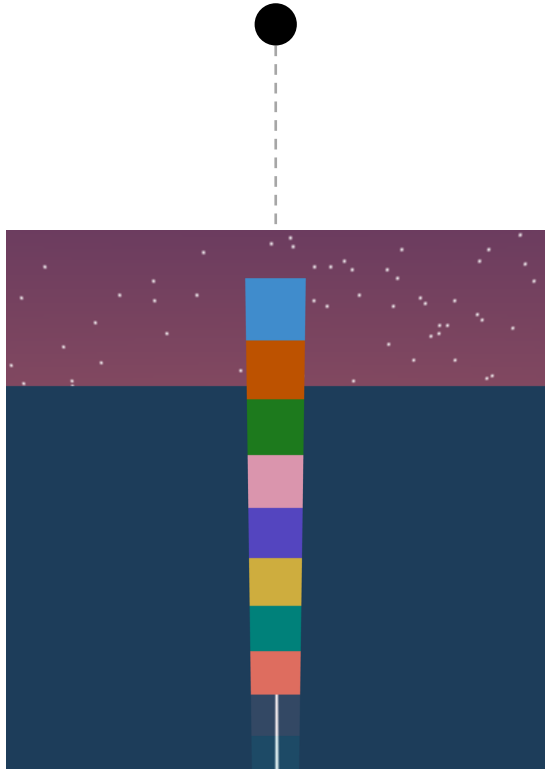


Initial state *Goal state*

What about Eight Blocks?

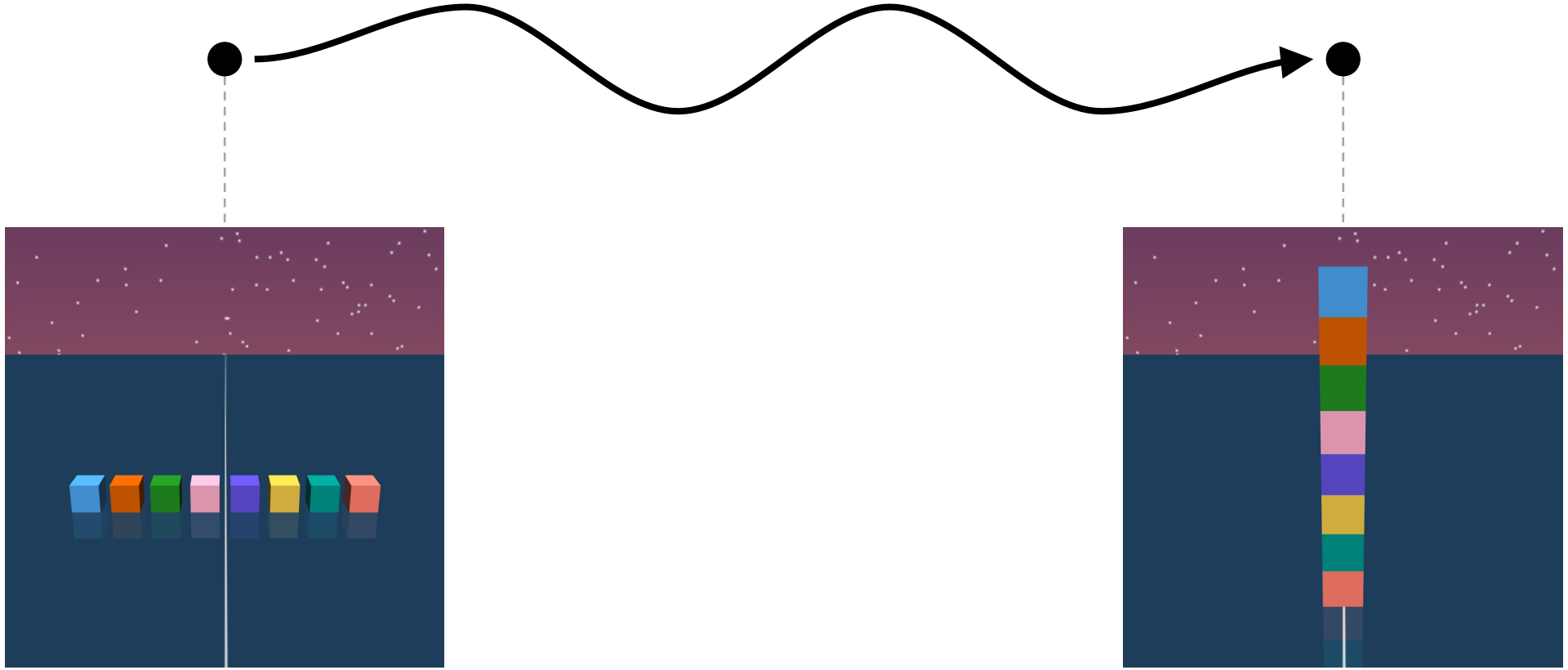


Initial state



Goal state

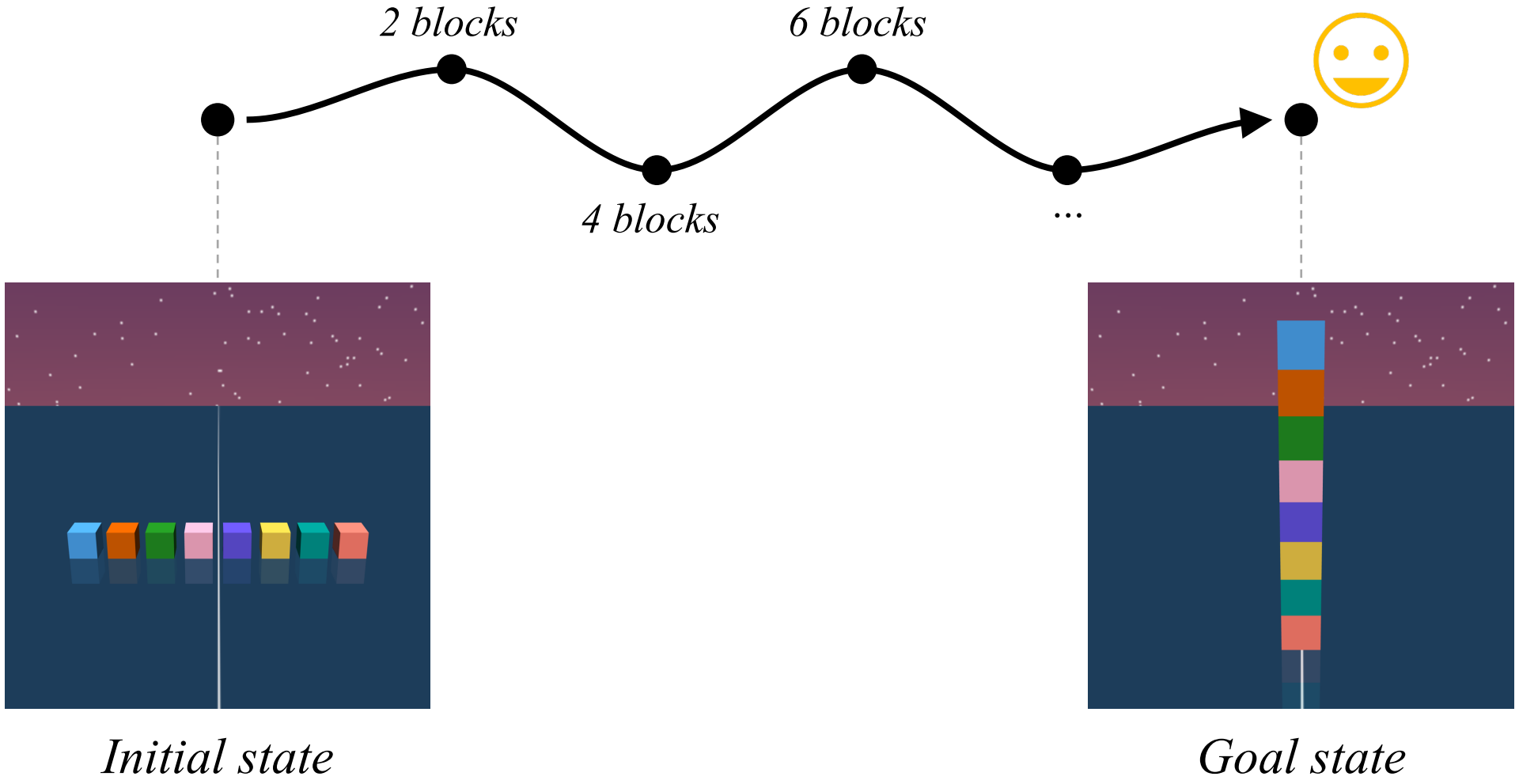
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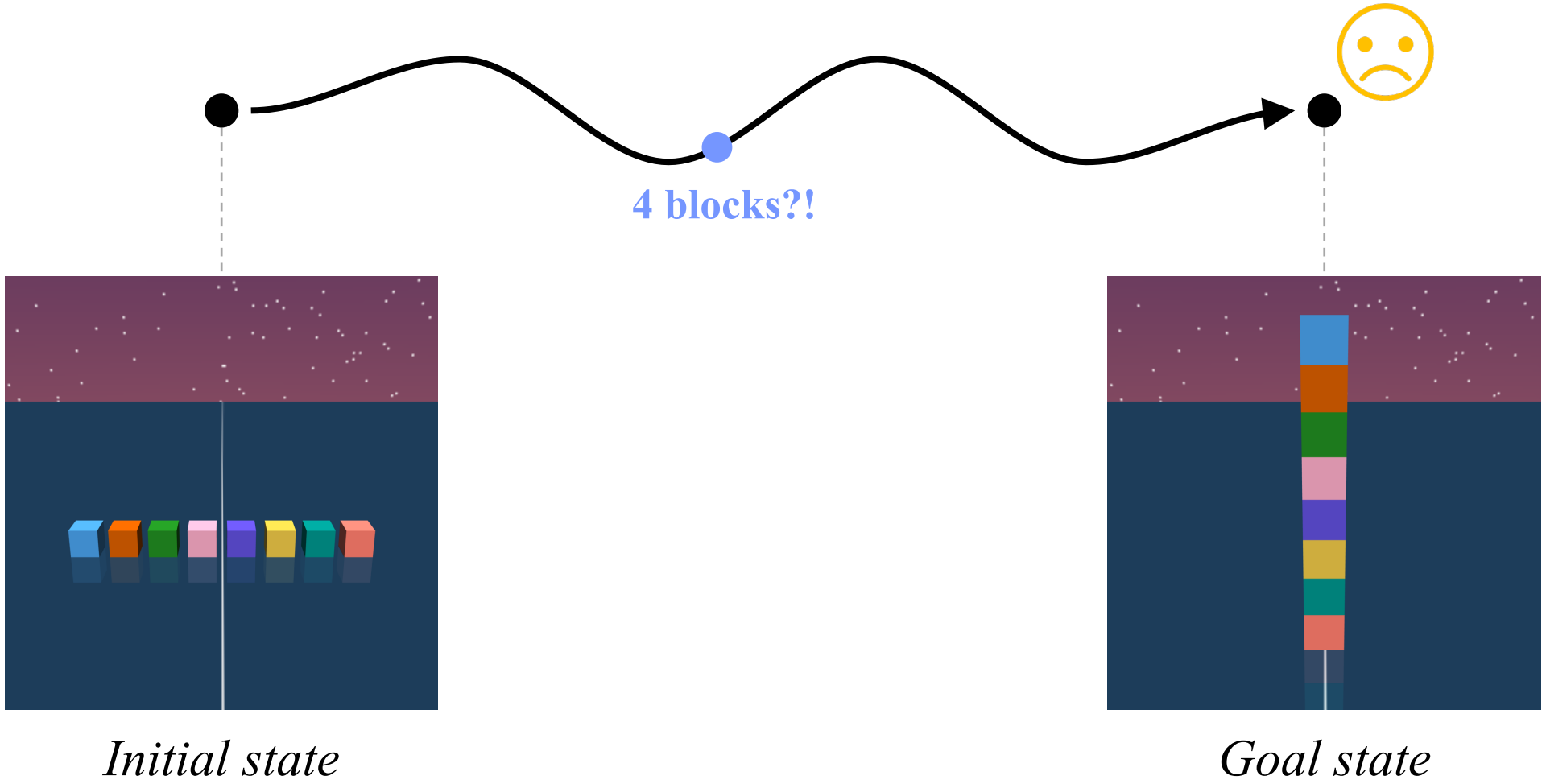
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Stacking Eight Blocks with too few planning stages



Desiderata for **Long-horizon** Planning and Exploration

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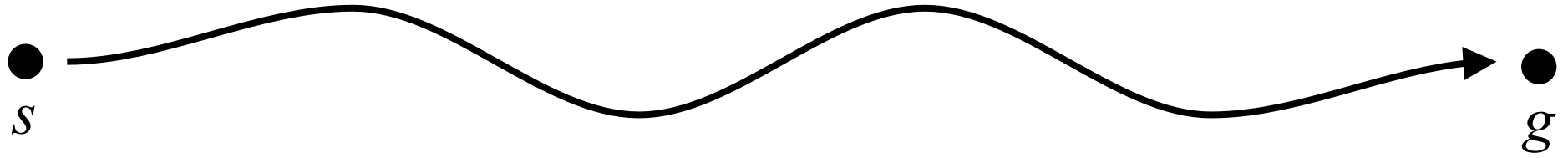
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($\approx$ *waypoints*)

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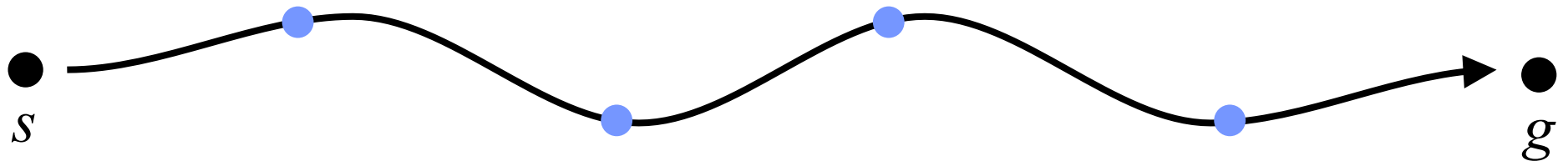
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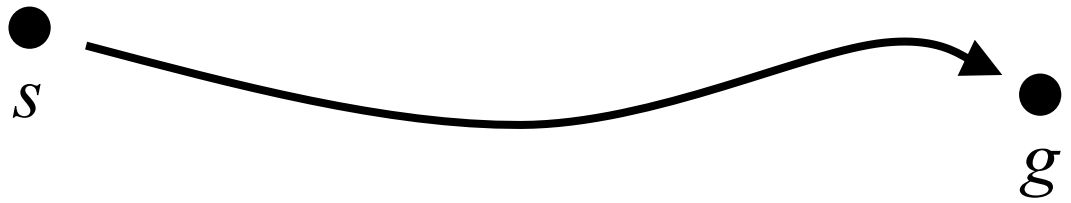


: Many waypoints

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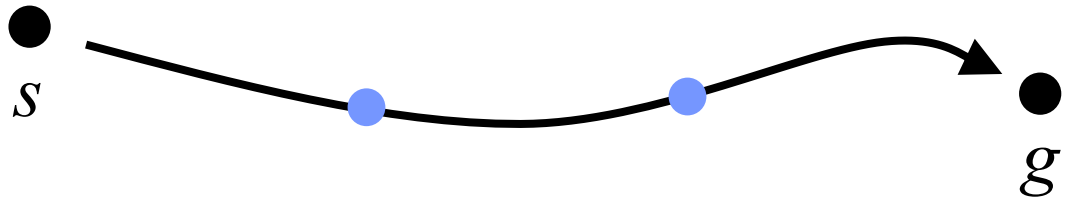
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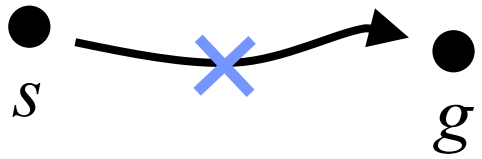


: Fewer waypoints

Desiderata for **Long-horizon** Planning and Exploration

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: Or even NONE

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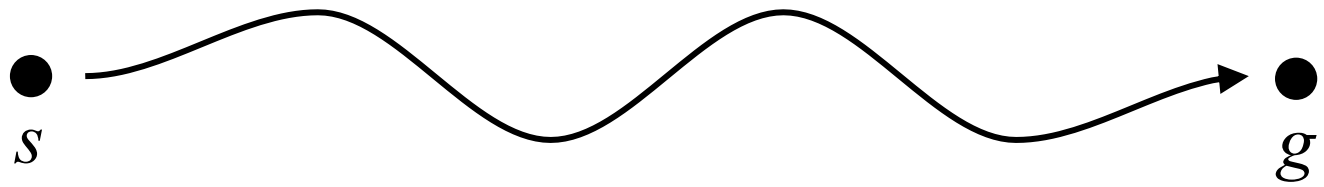
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Consider the eight-block stacking task,

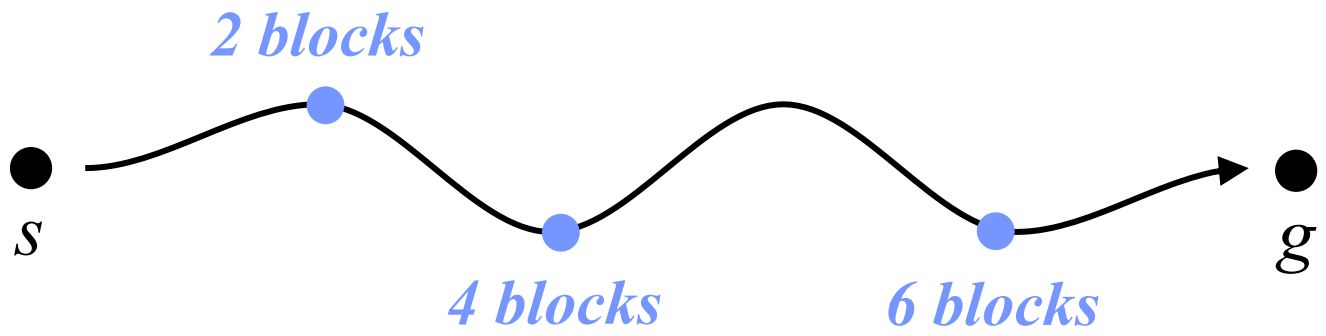


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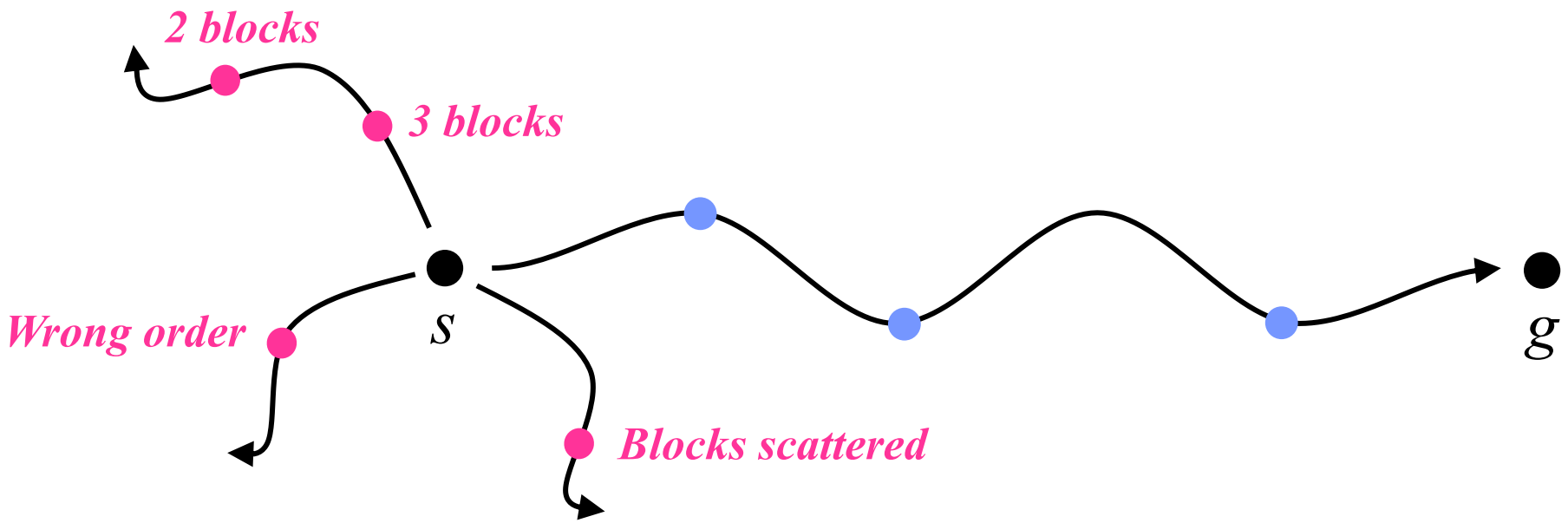
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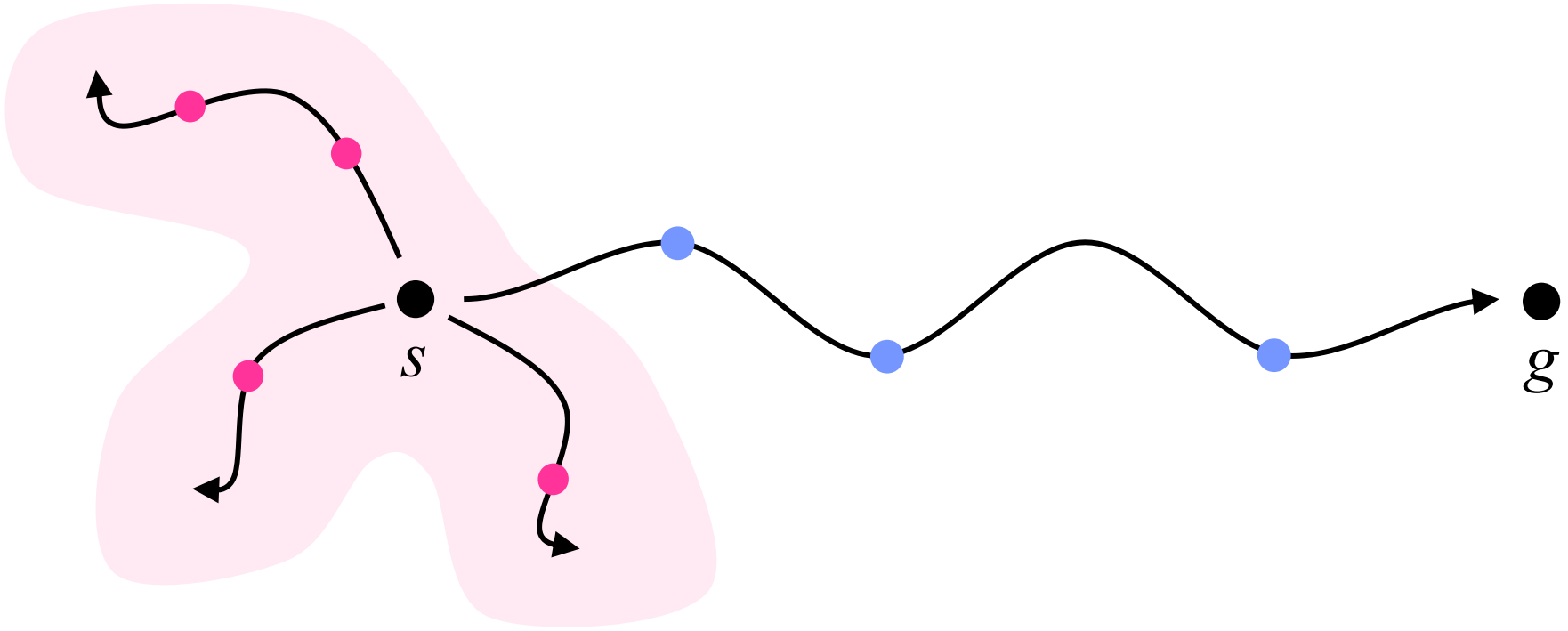
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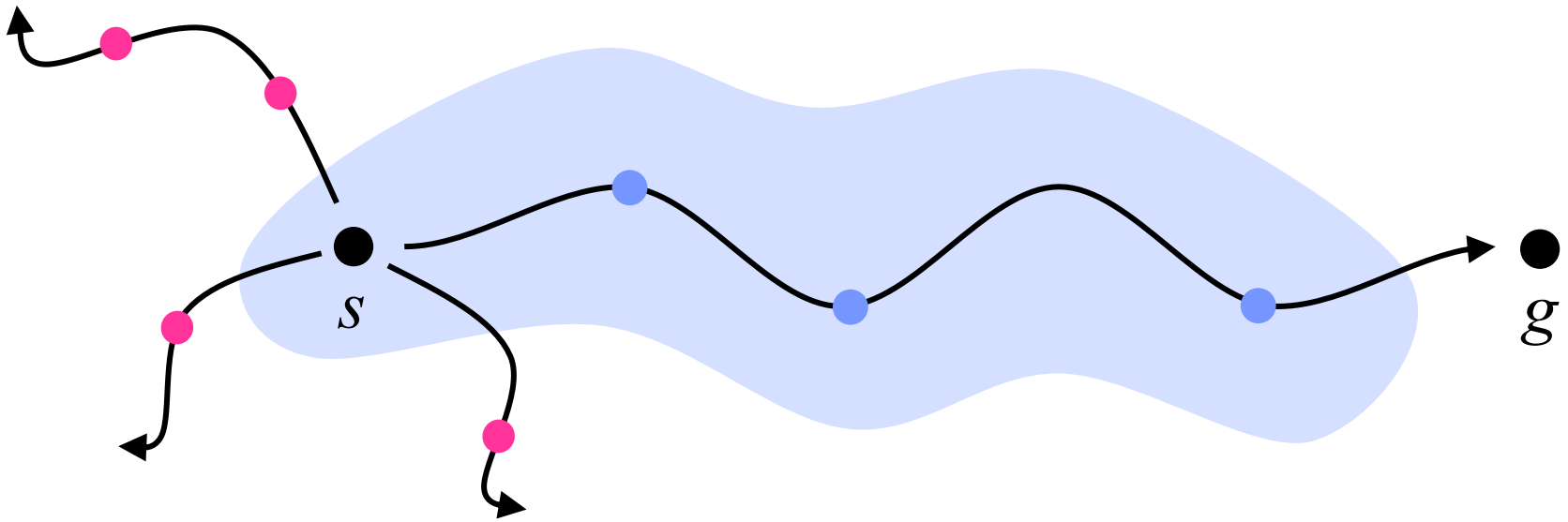
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Problem Setting 

Basic Setup : Data collection process

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High level planner

$z \sim \pi_H\ (z \mid s, g)$

(Low level) policy

$a \sim \pi_L\ (a \mid s, z)$



Basic Setup : Data collection process

Goal distribution

$\mathrm{g} \sim P_g(g)$

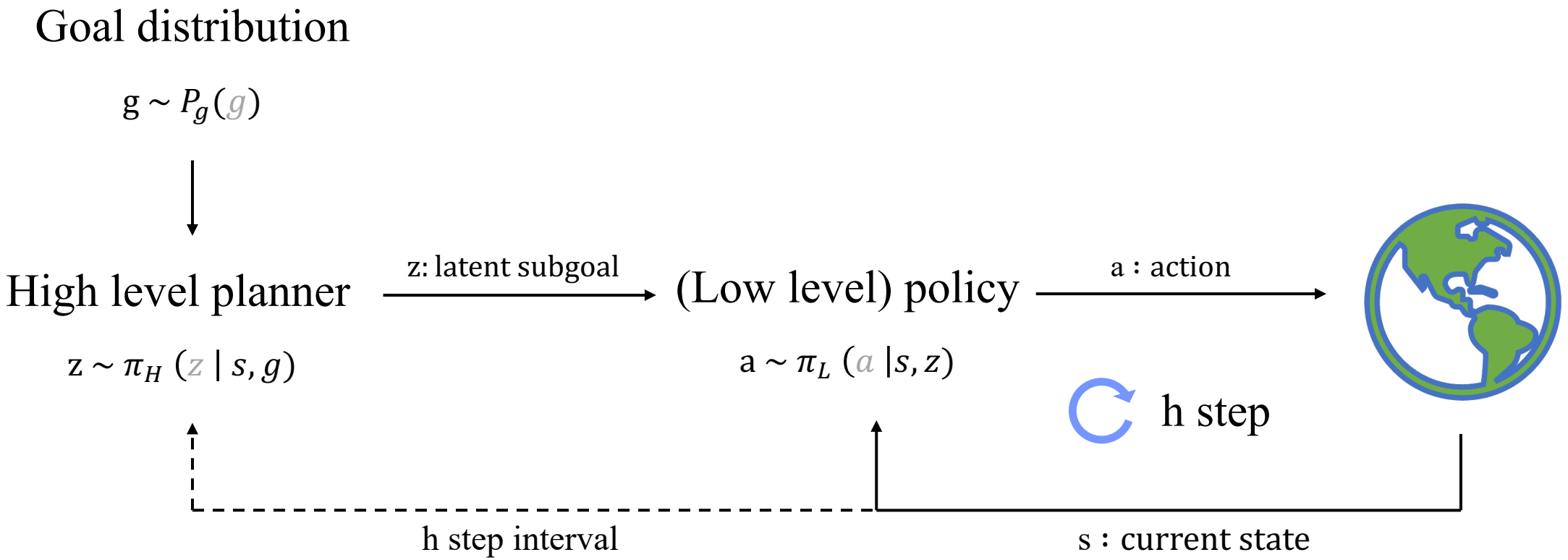


High level planner $\xrightarrow{\text{z: latent subgoal}}$ (Low level) policy

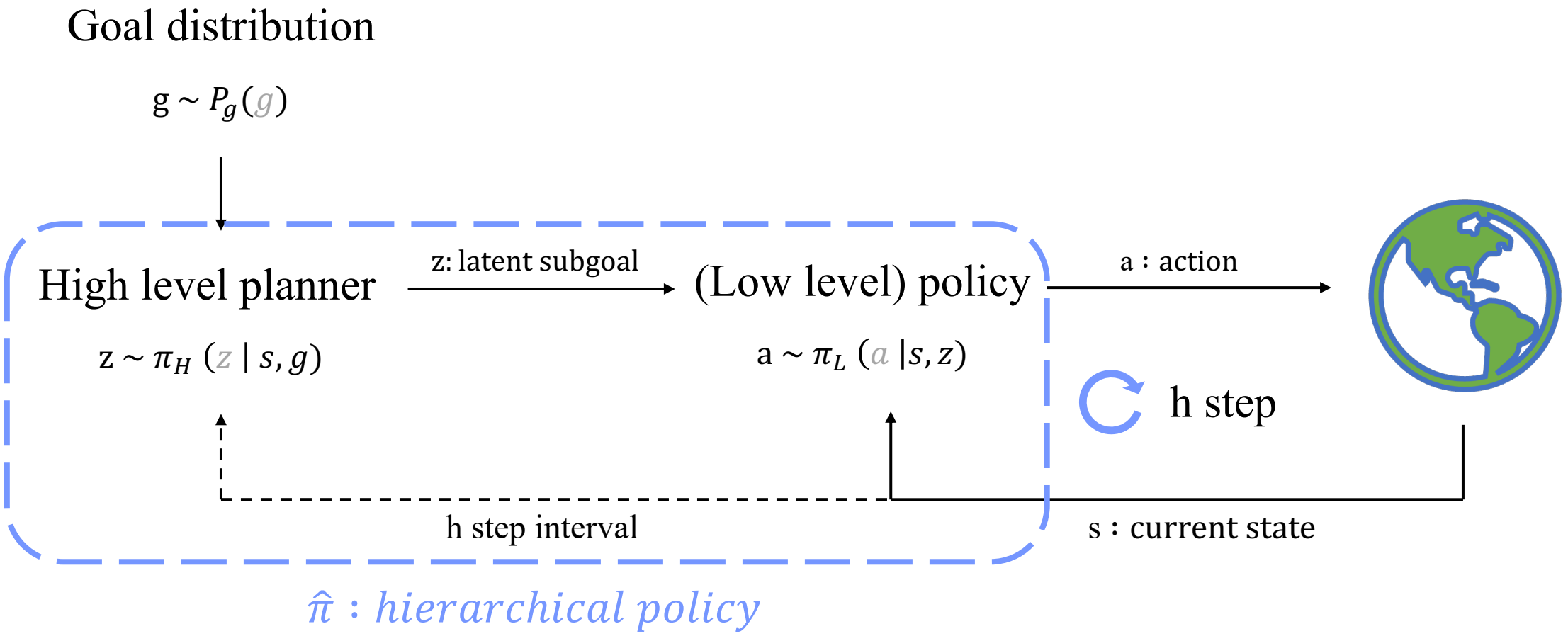
$\mathrm{z} \sim \pi_H\ (z \mid s, g)$ $\quad$ $\mathrm{a} \sim \pi_L\ (a \mid s, z)$



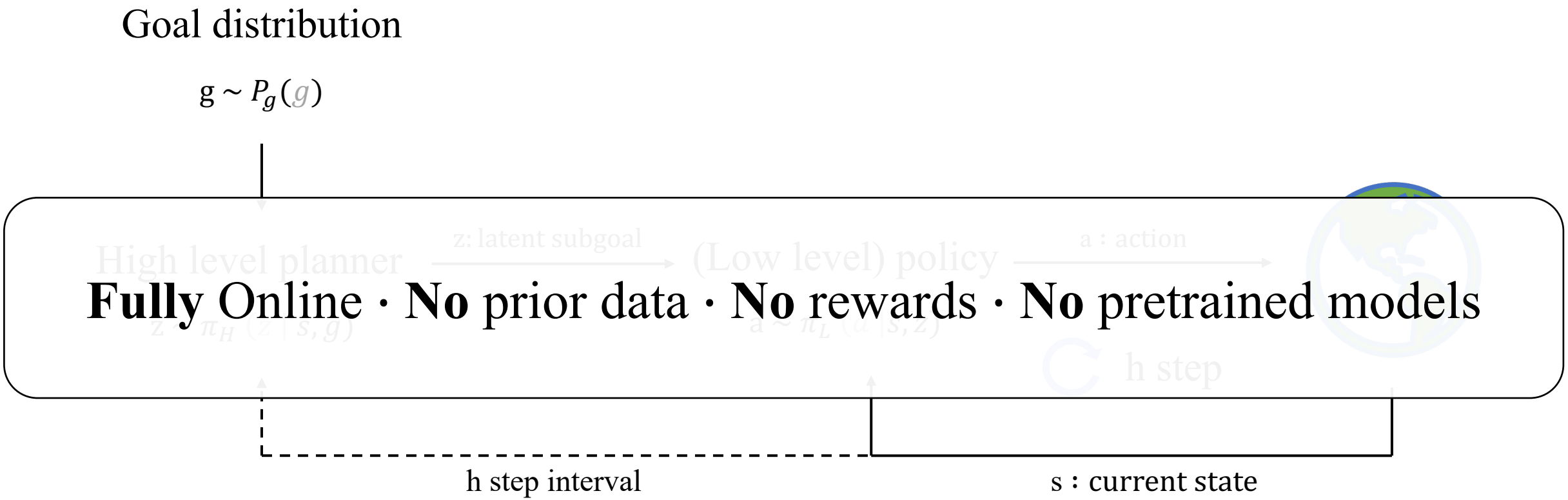
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Basic Setup : A probabilistic approach to planning

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Given a state *s* and a goal *g*,

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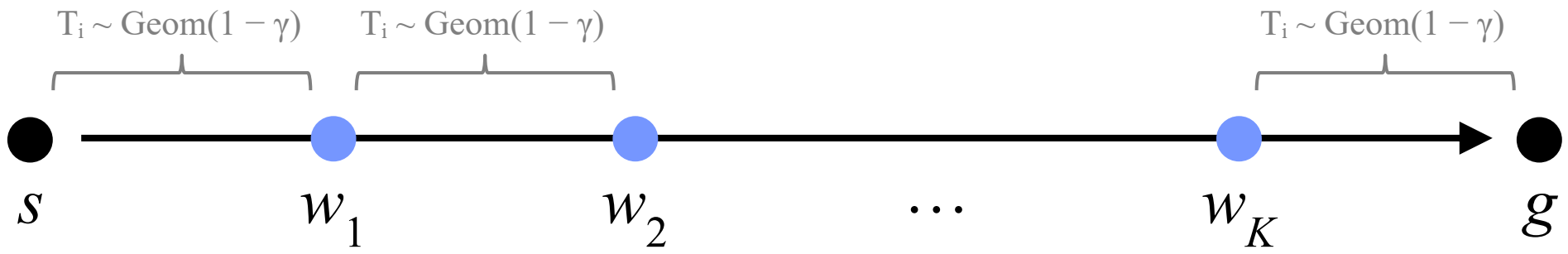
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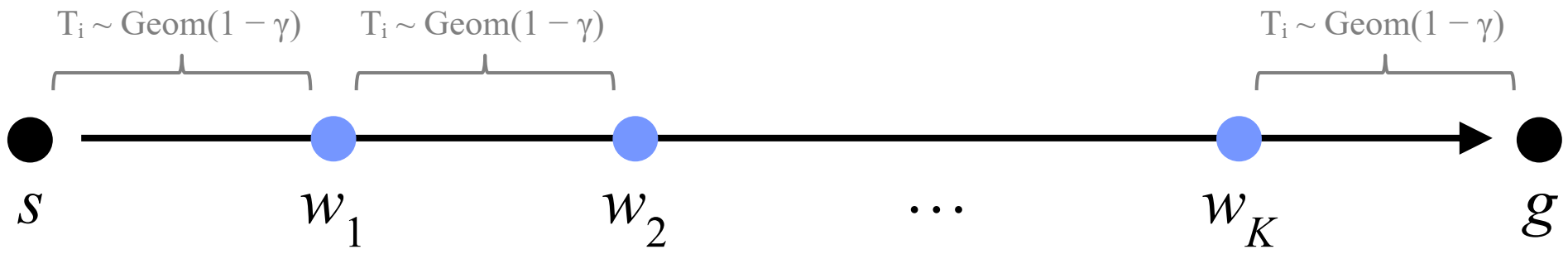
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For notational consistency, we set $w_0 = s$ and $w_{K+1} = g$

The Objective

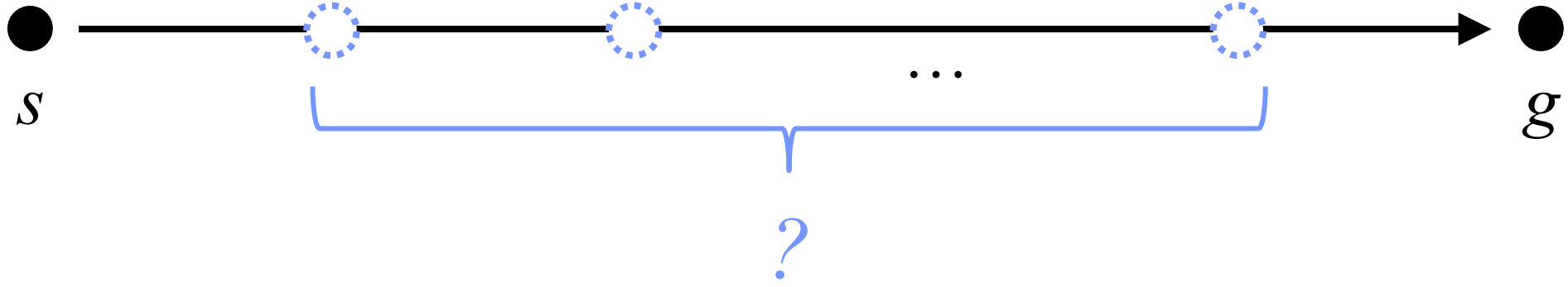
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Given *(s,g)* and number of waypoints *K*,



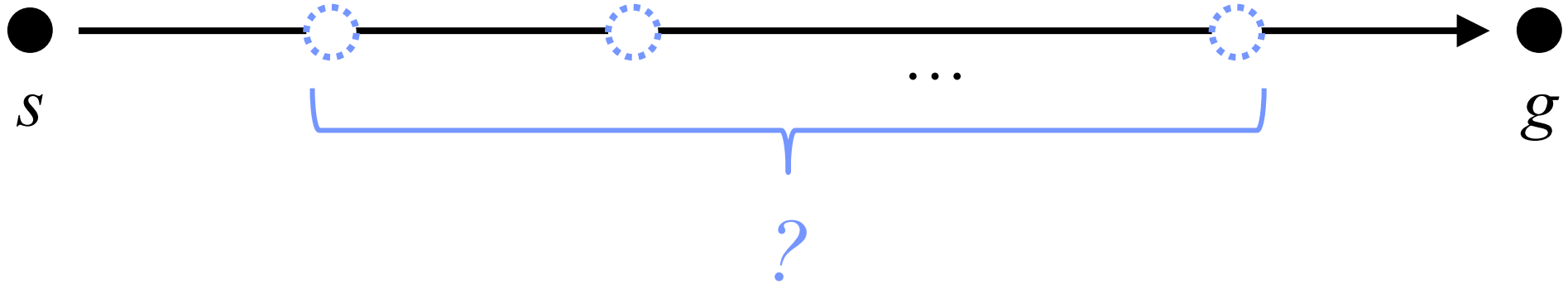
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For now, *K* is given. We then derive a **criterion for selecting *K* adaptively**.

The Objective

Given *(s,g)* and number of waypoints *K*,

$$q^\star = \arg\min_q D_{\mathrm{KL}}\Big(q(w_{1:K} \mid s, g) \;\Big\|\; p_{\hat{\pi}}^{\gamma,K}(w_{1:K} \mid s, g) \Big)$$

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The distribution over the ***K* intermediate waypoints** given s and g, under data collected by $\hat{\pi}$

The Objective

Given *(s,g)* and number of waypoints *K*,

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A tractable **approximation** to the posterior $p_{\hat{\pi}}^{\gamma,K}(\cdot|s,g)$

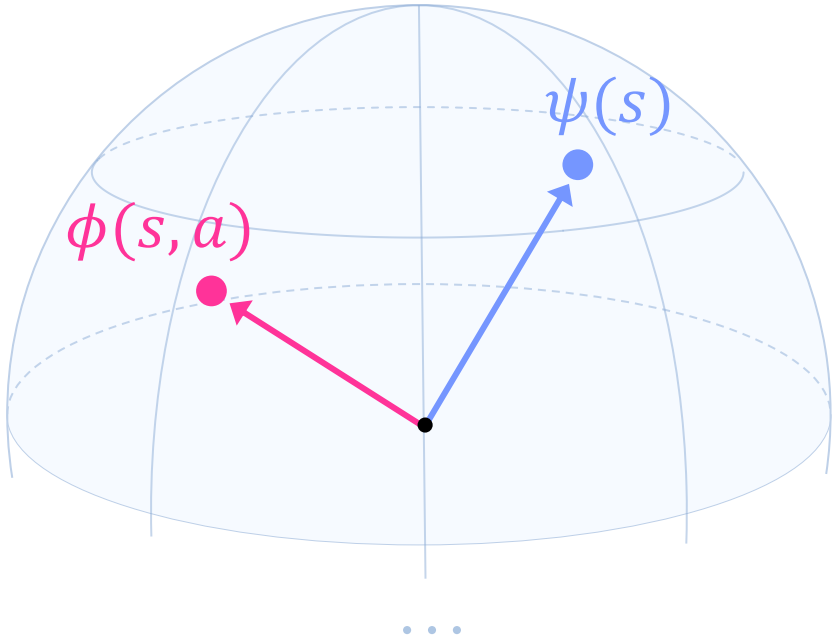
Hyperspherical Contrastive Representations

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Two representations on the unit hypersphere S^{d-1} + One learnable **rotation matrix** $\boldsymbol{R}$.

Hyperspherical Contrastive Representations

Two representations on the unit hypersphere $S^{\mathrm{d}-1}$ + One learnable **rotation matrix** R.



$\phi(s,a)$: state–action representation

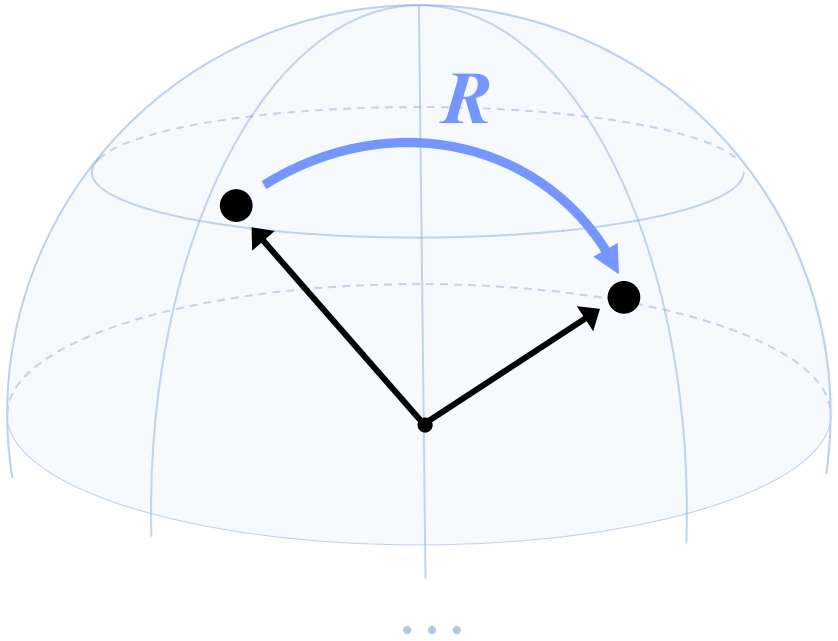
$\psi(s)$: state representation

$\|\phi(s,a)\|_2 = \|\psi(s)\|_2 = 1$

: lie on the unit hypersphere S^{d-1}

Hyperspherical Contrastive Representations

Two representations on the unit hypersphere S^{d-1} + One learnable **rotation matrix $\boldsymbol{R}$**.



Cayley transform of a *learnable* matrix A:

$$\left.\begin{aligned} \boldsymbol{\Omega} &= A - A^T \\ \boldsymbol{R} &= (I + \boldsymbol{\Omega})^{-1}(I - \boldsymbol{\Omega}) \end{aligned}\right\} \Rightarrow R^T R = I,\ \|Rx\| = \|x\|$$

: preserves hyperspherical geometry of representations

Parameterization

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1. Score for *planning*

$$f_H(s, s^+) = \psi(s)^T R_{learnable}\, \psi(s^+)$$

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1. ***Why use the same encoder for s and s+***?

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Parameterization

1. Score for *planning*

$$f_H(s, s^+) = \psi(s)^T R_{learnable}\, \psi(s^+)$$

1. ***Why use the same encoder for s and s+?***

   To infer consecutive waypoints in a common latent space.

2. ***Why introduce the rotation matrix R ?***

   In general, dynamics are *asymmetric*,

   but the inner product is *symmetric* : $\psi(s)^T\psi(s^+) = \psi(s^+)^T\psi(s)$

Parameterization

1. Score for *planning*

$$f_H(s, s^+) = \psi(s)^T R_{learnable}\, \psi(s^+)$$

2. Score for *training policy*

$$f_L(s, a, s^+) = \phi(s, a)^T \psi(s^+)$$

Representation Learning

Sample a batch $\mathcal{B} = \{(s_i, a_i, s_i^+)\}_{i=1}^B$ from the replay buffer, where s_i^+ is $\Delta \sim \text{Geom}(1-\gamma)$ steps after s_i:

$$\mathcal{L}^{symmetric}(R, \psi; \mathcal{B}) = -\frac{1}{B}\sum_{i=1}^{B}\left[\log \frac{\exp(f_H(s_i, s_i^+))}{\sum_{j=1}^{B}\exp(f_H(s_i, s_j^+))} + \log \frac{\exp(f_H(s_i, s_i^+))}{\sum_{j=1}^{B}\exp(f_H(s_j, s_i^+))}\right]$$

$$\mathcal{L}^{backward}(\phi, \psi; \mathcal{B}) = -\frac{1}{B}\sum_{i=1}^{B}\log \frac{\exp(f_L(s_i, a_i, s_i^+))}{\sum_{j=1}^{B}\exp(f_L(s_j, a_j, s_i^+))}$$

$$\mathcal{L}_{repr} = E_{\mathcal{B}}\left[\mathcal{L}^{\text{symmetric}}(R, \psi; \mathcal{B}) + \lambda_{repr} \cdot \mathcal{L}^{\text{backward}}(\phi, \psi; \mathcal{B})\right]$$

Policy Learning

$$\max_{\theta} \mathbb{E}_{s\sim p_{\mathcal{D}},\, g\sim p_{\mathcal{D}}^{\gamma}(\cdot|s),\, a\sim\pi_{\theta}(\cdot \mid s,\psi(g))} \left[\phi(s,a)^{T}\psi(g)\right]$$

Policy Learning

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Policy Learning

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Same as CRL [1], except for this

[1] Eysenbach, Benjamin, et al. "Contrastive learning as goal-conditioned reinforcement learning." *2022.*

Key Assumptions on the Representations

[Assumption 1] The learned f_H under the symmetric InfoNCE objective has the following *Bayes-optimal* form:

$$f_H(u, v) = \log \frac{p^{\gamma}_{\widehat{\pi}}(u^+ = v \mid u)}{p^{\gamma}_{\widehat{\pi}}(v)\, C} \quad [1]$$

[Assumption 2] The marginal distribution over the representations ψ is *uniform* on the unit hypersphere $\mathbb{S}^{d-1}$:

$$p^{\gamma}_{\widehat{\pi}}(\psi) = \mathrm{Unif}\left(\mathbb{S}^{d-1}\right) \quad [2]$$

[1] Eysenbach, Benjamin, et al. "Inference via interpolation: Contrastive representations provably enable planning and inference." *2024.*
[2] Wang, Tongzhou, et al. "Understanding contrastive representation learning through alignment and uniformity on the hypersphere." *2020.*

Tractable Reformulation

#1

$$\arg\min_{q} D_{\mathrm{KL}}\left(q(w_{1:K} \mid s, g) \,\middle\|\, p_{\hat{\pi}}^{\gamma, K}(w_{1:K} \mid s, g)\right)$$

Assumption 1 & Markov assumption

$$q(w_{1:K} \mid s, g) = \prod_{i=1}^{K} q_i(w_i \mid s, g)$$

#2

$$\arg\max_{q}\left\{\mathbb{E}_{w_{1:K}\sim q(w_{1:K}|s,g)}\left[\sum_{i=0}^{K} f_H(w_i, w_{i+1})\right] - \sum_{i=1}^{K} D_{\mathrm{KL}}\left(q_i(w_i \mid s, g) \,\|\, p_{\hat{\pi}}^{\gamma}(w_i)\right)\right\}$$

Tractable Reformulation

#2

$$\arg\max_{q} \left\{ \mathbb{E}_{w_{1:K} \sim q(w_{1:K}|s,g)} \left[\sum_{i=0}^{K} f_H(w_i, w_{i+1}) \right] - \sum_{i=1}^{K} D_{\mathrm{KL}} \left(q_i(w_i \mid s, g) \parallel p^{\gamma}_{\hat{\pi}}(w_i) \right) \right\}$$

Assumption 2 & $f_H(w_i, w_{i+1}) = \psi(w_i) R \psi(w_{i+1})$

$\psi(w_i) \rightarrow z_i$

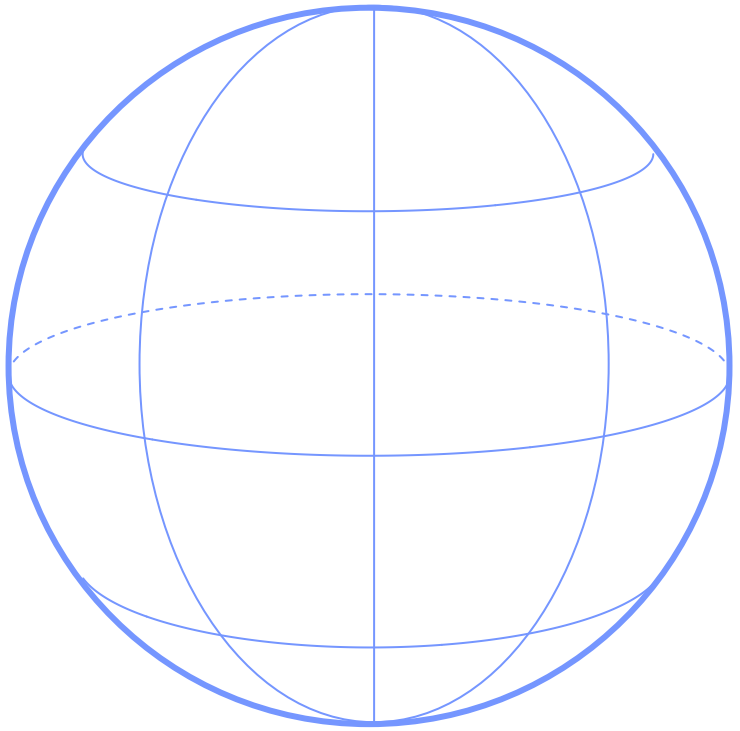
$q_i(w_i|s,g) \rightarrow q_i(z_i|s,g)$

#3

$$\arg\max_{q} \mathbb{E}_{z_{1:K} \sim q(z_{1:K}|s,g)} \left[\psi(s)^{\top} R z_1 + \sum_{i=2}^{K} z_{i-1}^{\top} R z_i + z_K^{\top} R \psi(g) \right] + \sum_{i=1}^{K} \mathcal{H} \left(q_i(z_i \mid s, g) \right)$$

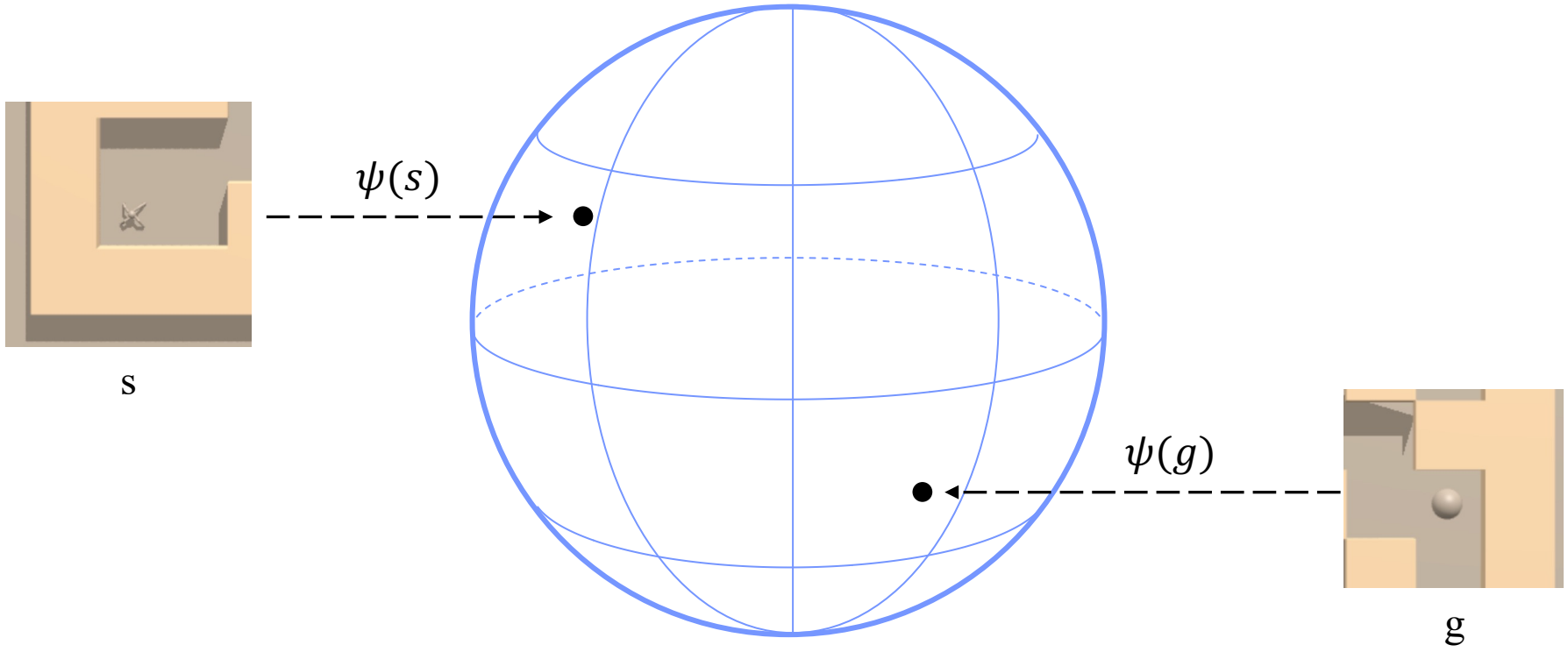
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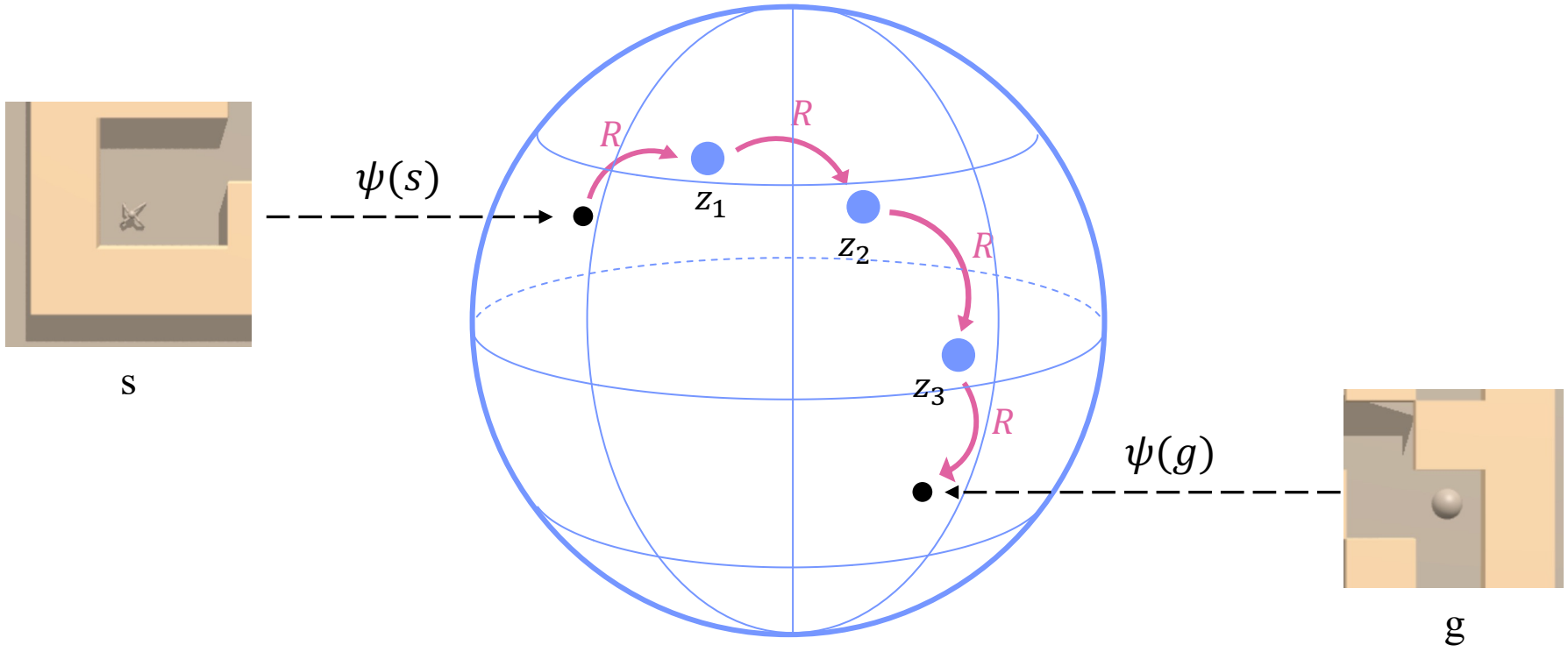
$$\arg\max_{q} \mathbb{E}_{z_{1:K}\sim q(z_{1:K}|s,g)}\left[\psi(s)^\top R z_1 + \sum_{i=2}^{K} z_{i-1}^\top R z_i + z_K^\top R\psi(g)\right] + \sum_{i=1}^{K}\mathcal{H}\left(q_i(z_i \mid s,g)\right)$$

Geometric Interpretation



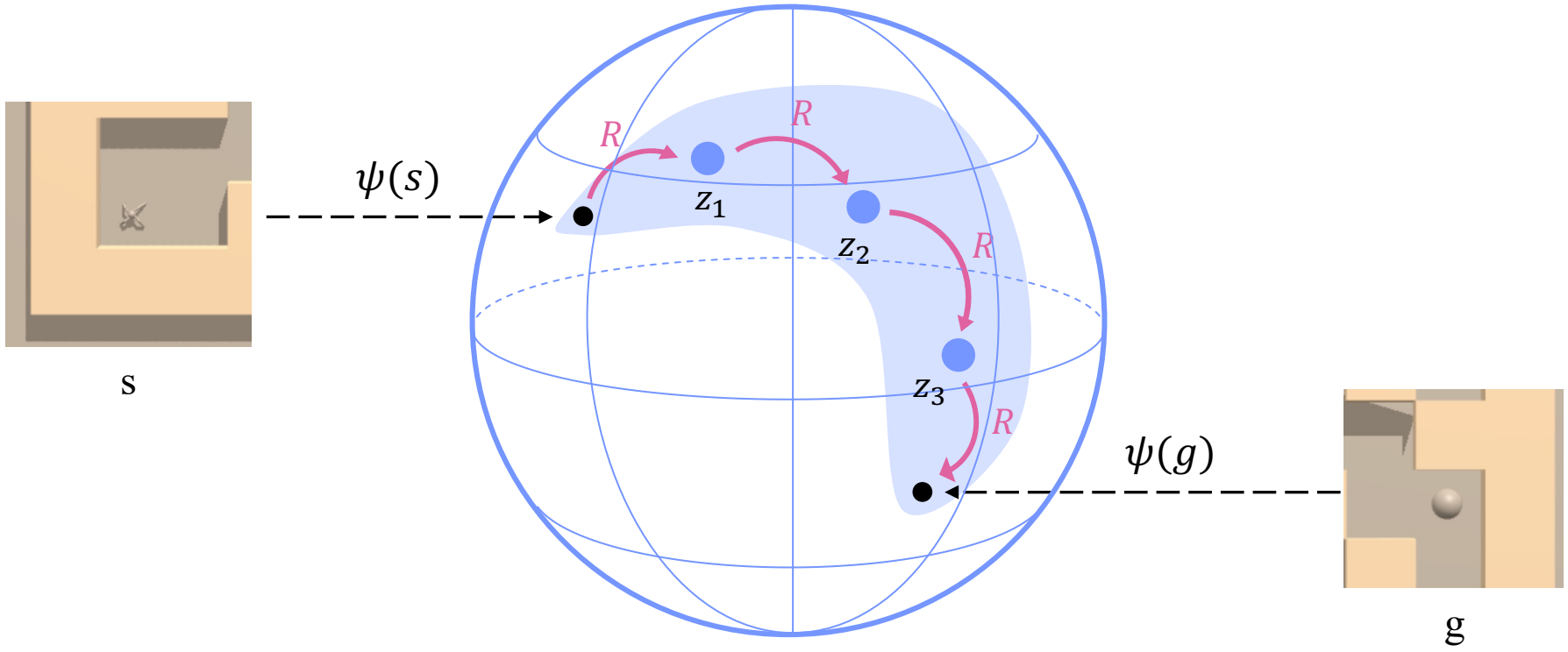
$$\arg\max_{q} \mathbb{E}_{z_{1:K}\sim q(z_{1:K}|s,g)} \left[\psi(s)^\top R z_1 + \sum_{i=2}^{K} z_{i-1}^\top R z_i + z_K^\top R \psi(g) \right] + \sum_{i=1}^{K} \mathcal{H}\left(q_i(z_i \mid s, g)\right)$$

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Parameterize q_i by normalizing **Gaussian samples** onto the unit **hypersphere** :

$$\tilde{z}_i \sim \mathcal{N}\left(\mu_i(s,g),\, \sigma_i^2(s,g)I\right), \quad z_i = \frac{\tilde{z}_i}{\|\tilde{z}_i\|_2} \sim q_i(\cdot \mid s,g)$$

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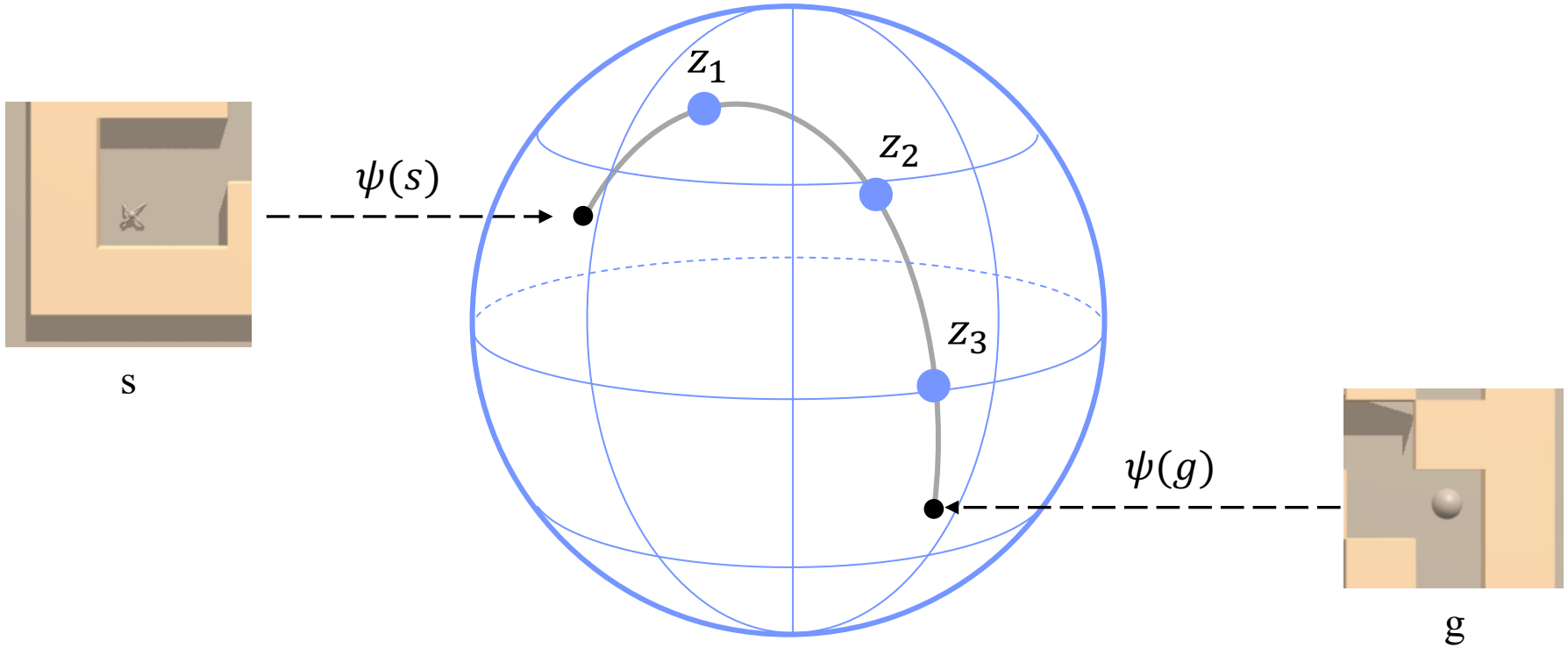
Under the small-variance assumption,

$$\begin{cases} \mu_i^*(s,g) = (R^T)^i \,\mathrm{slerp}\Big(\psi(s),\, R^{K+1}\psi(g),\, \frac{i}{K+1}\Big), \\ \sigma_i^{*2}(s,g) = \dfrac{\alpha}{2\cos(\frac{\theta}{K+1})} \end{cases} \qquad \text{for } i = 1,\ldots,K,$$

$^*\theta = \arccos(\psi(s)^\top R^{K+1}\psi(g))$

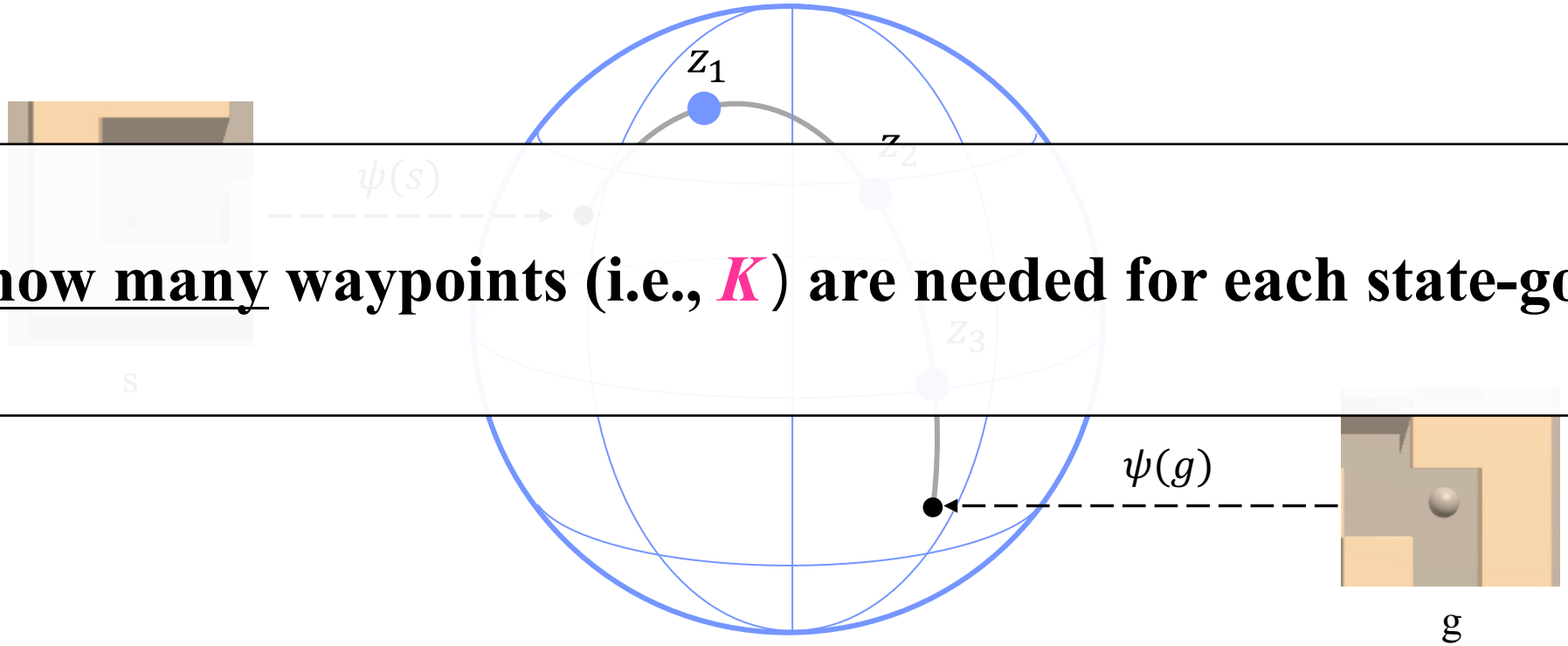
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Then, <u>how many</u> waypoints (i.e., *K*) are needed for each state-goal pair?

K–selection strategy

K-selection strategy

$$f_H(s, s^+) = \psi(s)^\top R \psi(s^+) = (R^\top \psi(s))^\top \psi(s^+)$$: one geometric transition

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K-selection strategy (see Appendix A.4 for the detailed derivation)

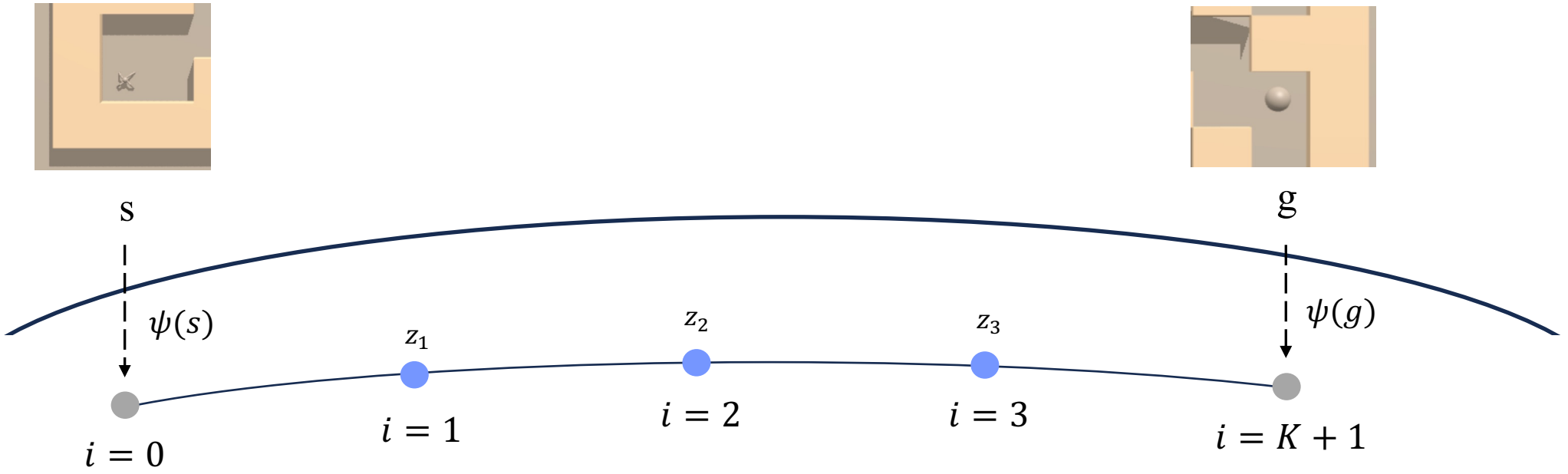
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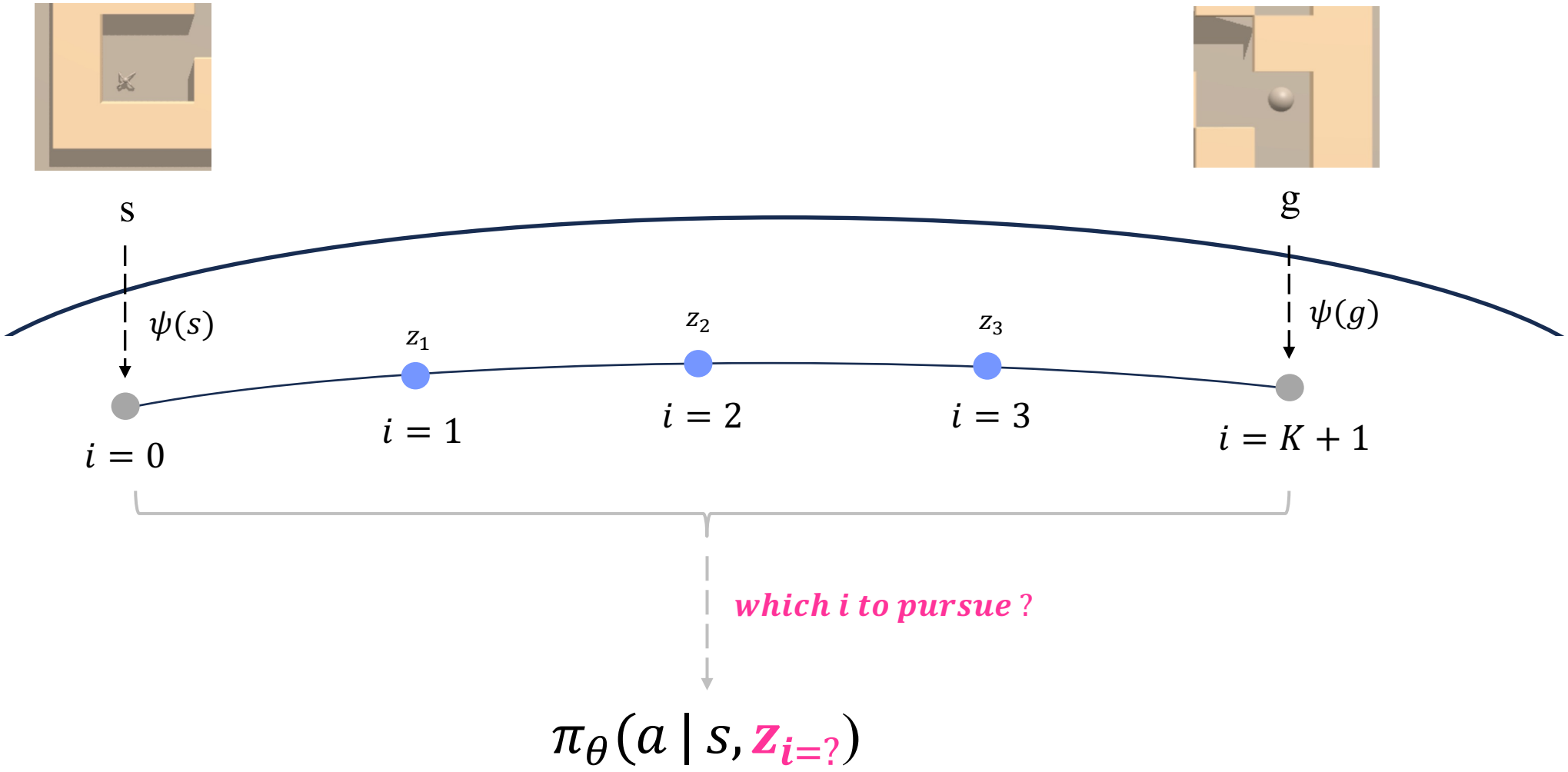
↓ vMF likelihood & Bayes' rule

$$\hat{p}(K \mid \psi(s), \psi(g)) = \mathrm{Softmax}_{K \in \{0, \ldots, K_{\max}\}} \left(\frac{\ell_K(s, g)}{\eta} \right)$$

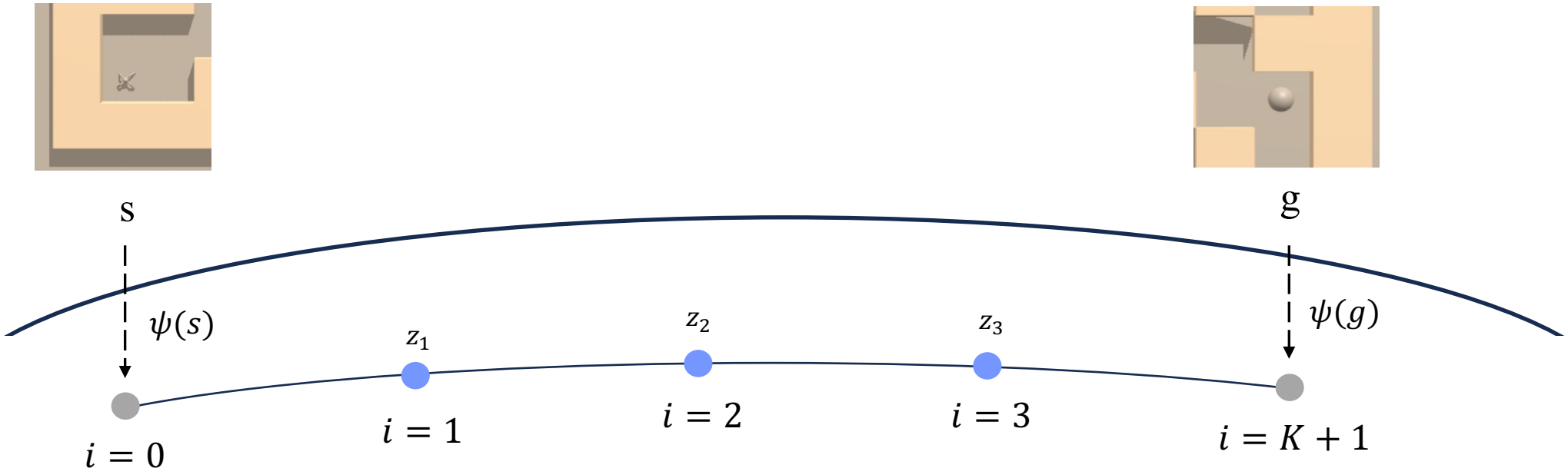
i – selection strategy



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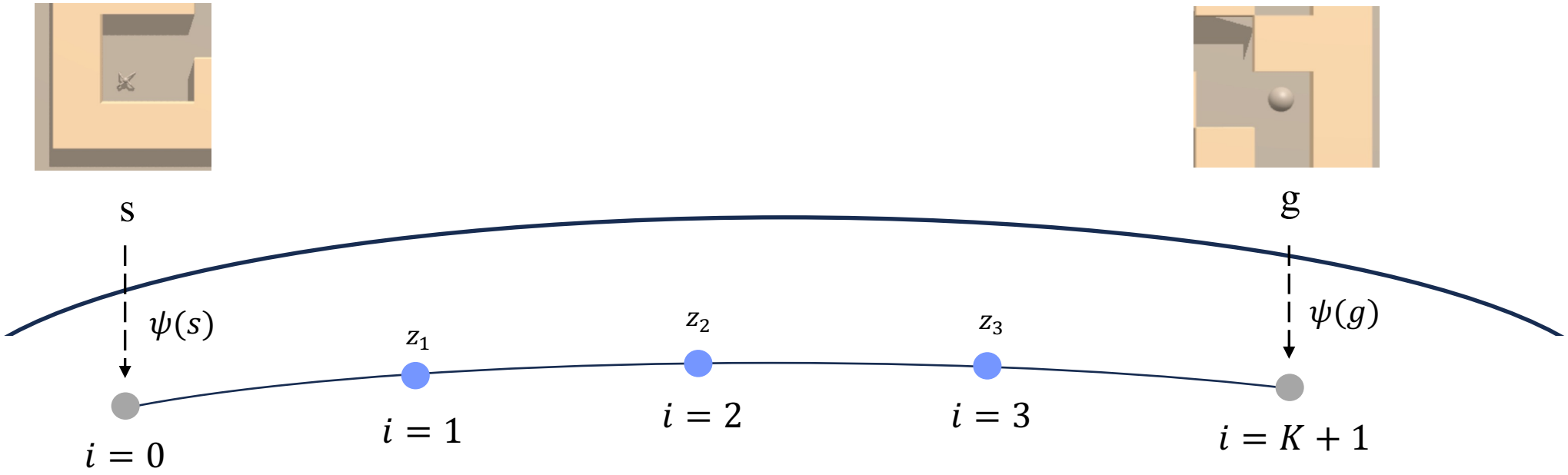


i – selection strategy



Are subgoals reachable by the low-level policy <u>for all *i*</u> ? 🤔

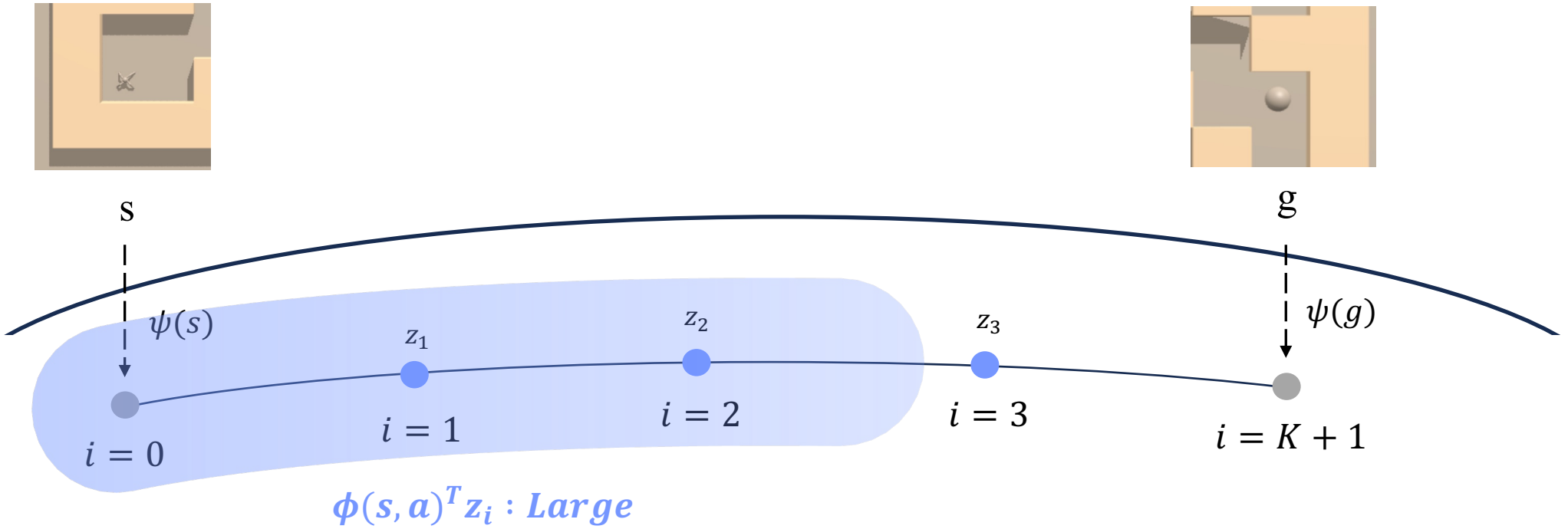
Proxy for reachability



- Recall the policy learning objective

$$\max_{\theta} \mathbb{E}_{s\sim p_{\mathcal{D}},\, w\sim p_{\mathcal{D}}^{\gamma}(\cdot|s),\, a\sim\pi_{\theta}(\cdot\,|\,s,\psi(w))} \left[\phi(s,a)^{T} \underbrace{\psi(w)}_{\boldsymbol{z_i}}\right]$$

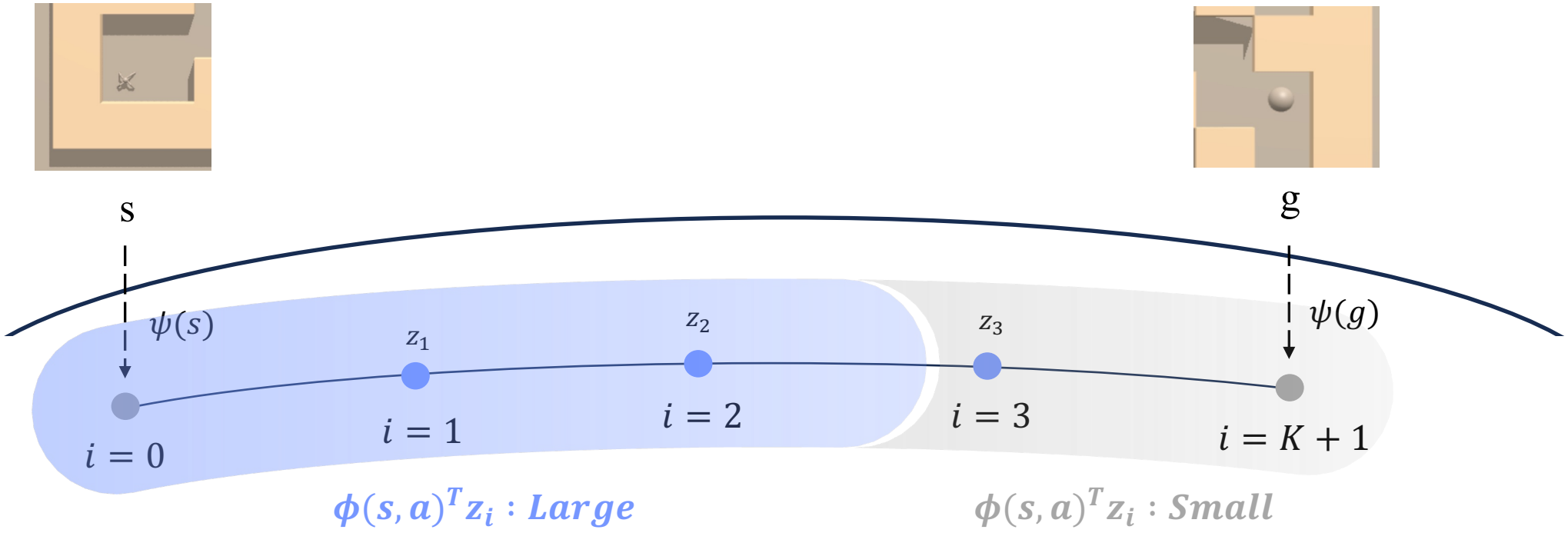
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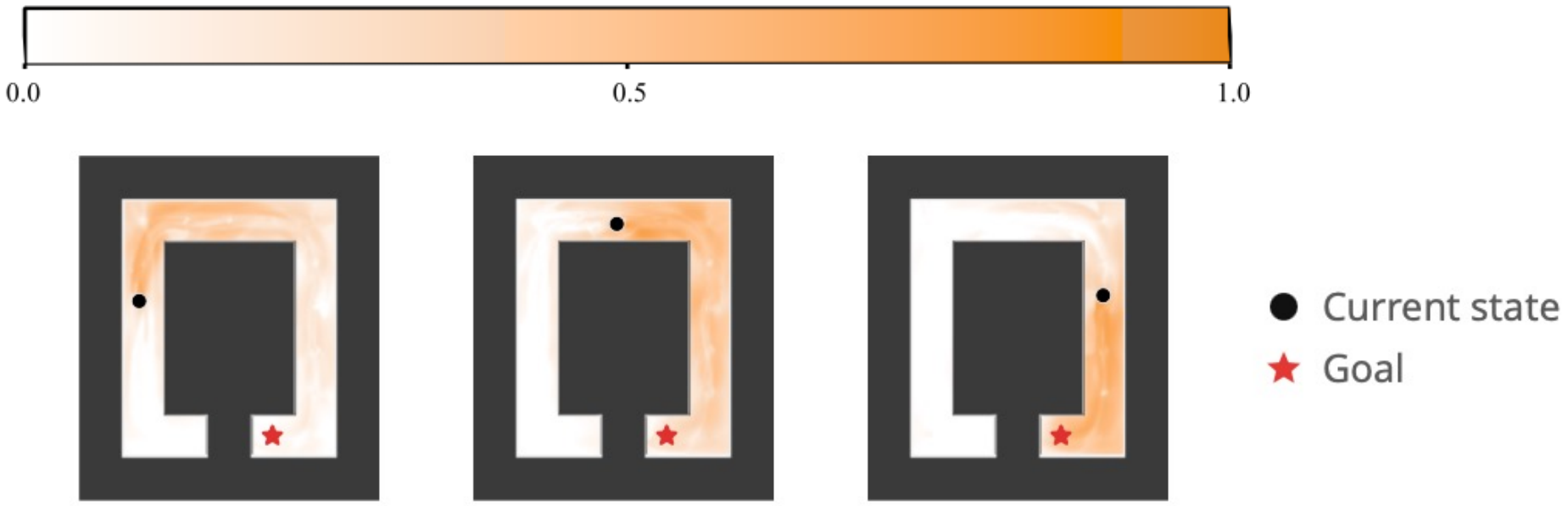


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Low level reachability : visualization

$$Low\ level\ reachability\ score: \ f_L(s,a,w) = \phi(s,a)^T \underbrace{\psi(w)}_{\boldsymbol{z_i}}$$



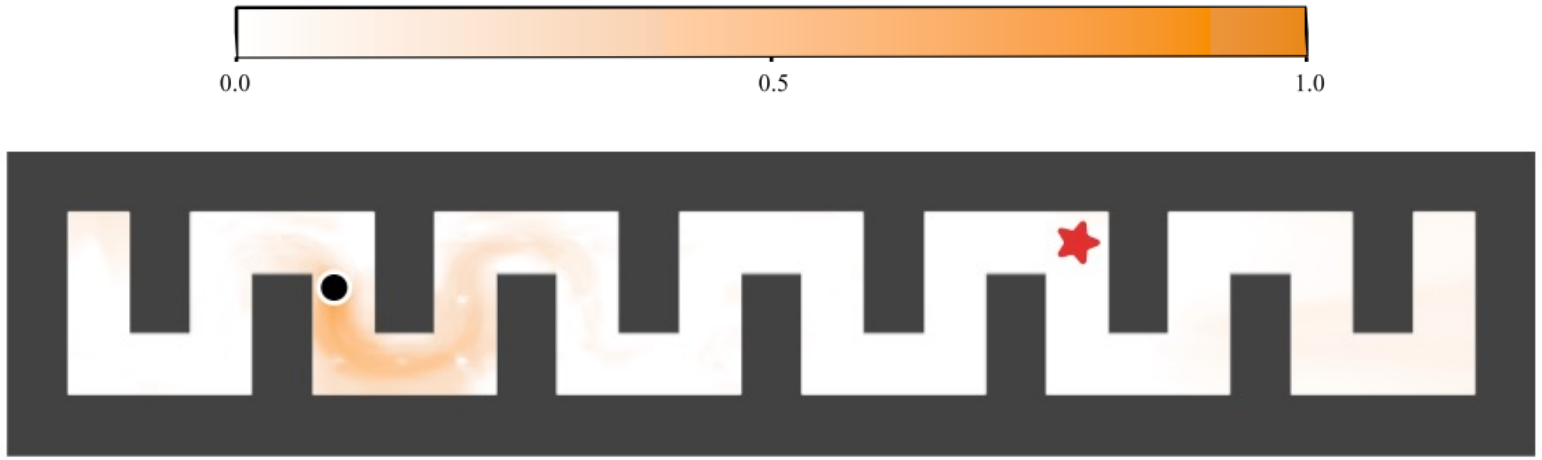
Higher **score** indicates greater estimated reachability of the current policy

i – selection strategy

i – selection strategy

1. Compute the $\boldsymbol{f_L}$ **score** for each candidate waypoint representation z_i

$$\rho_i = \phi(s, a_i)^\top z_i, \; a_i \sim \pi_\theta(\cdot \mid s, z_i)$$

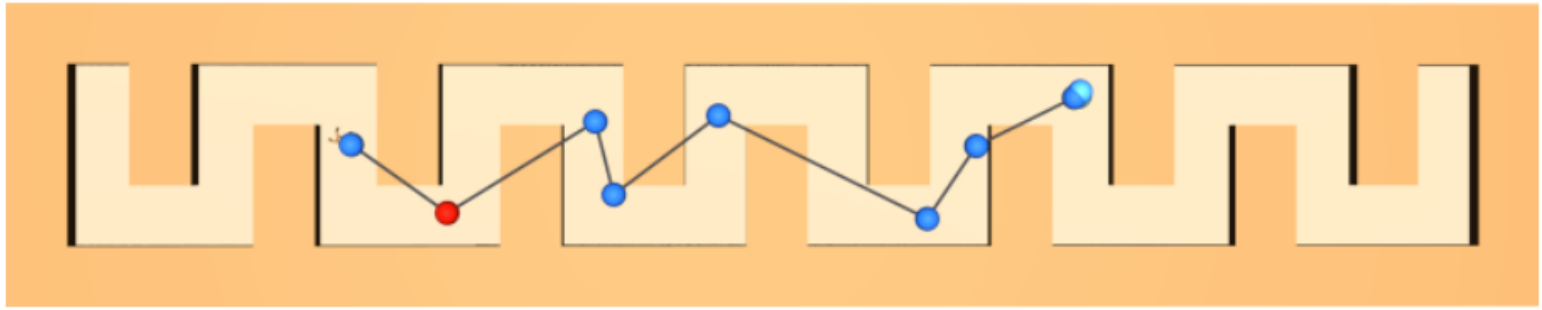


i – selection strategy

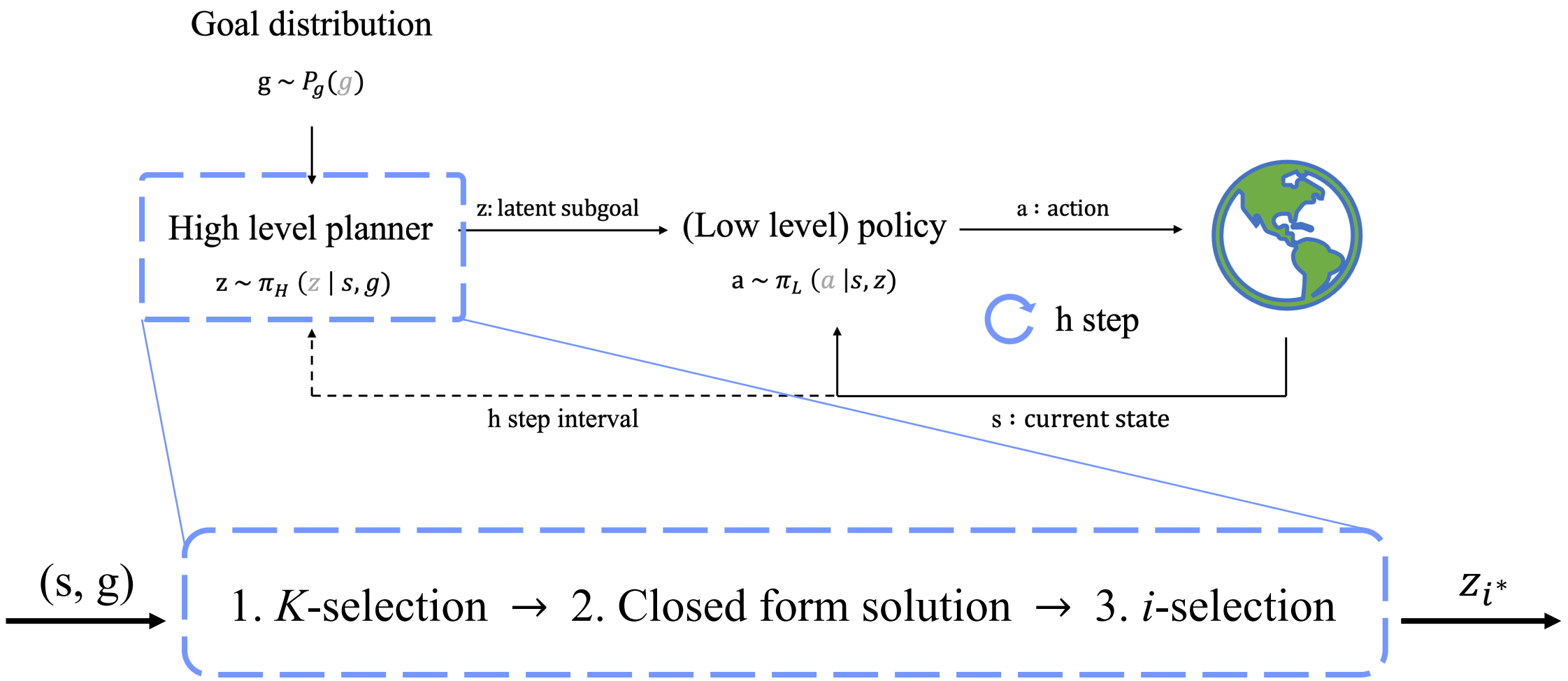
1. Compute the f_L **score** for each candidate waypoint representation z_i

2. Select the **furthest waypoint** whose score meets the threshold ϵ

$$i^* = \max\{i \in \{1, \ldots, K+1\} : \rho_i \geq \epsilon\}$$

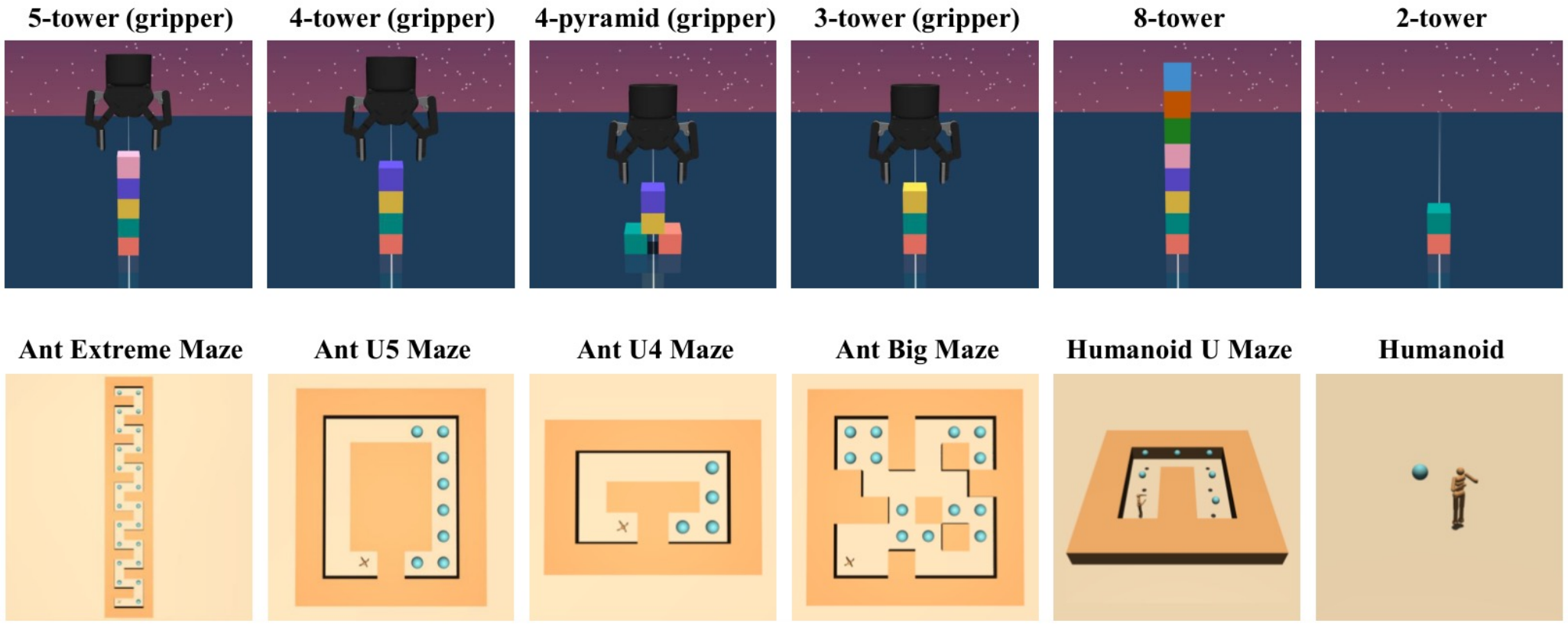


Overall planning algorithm



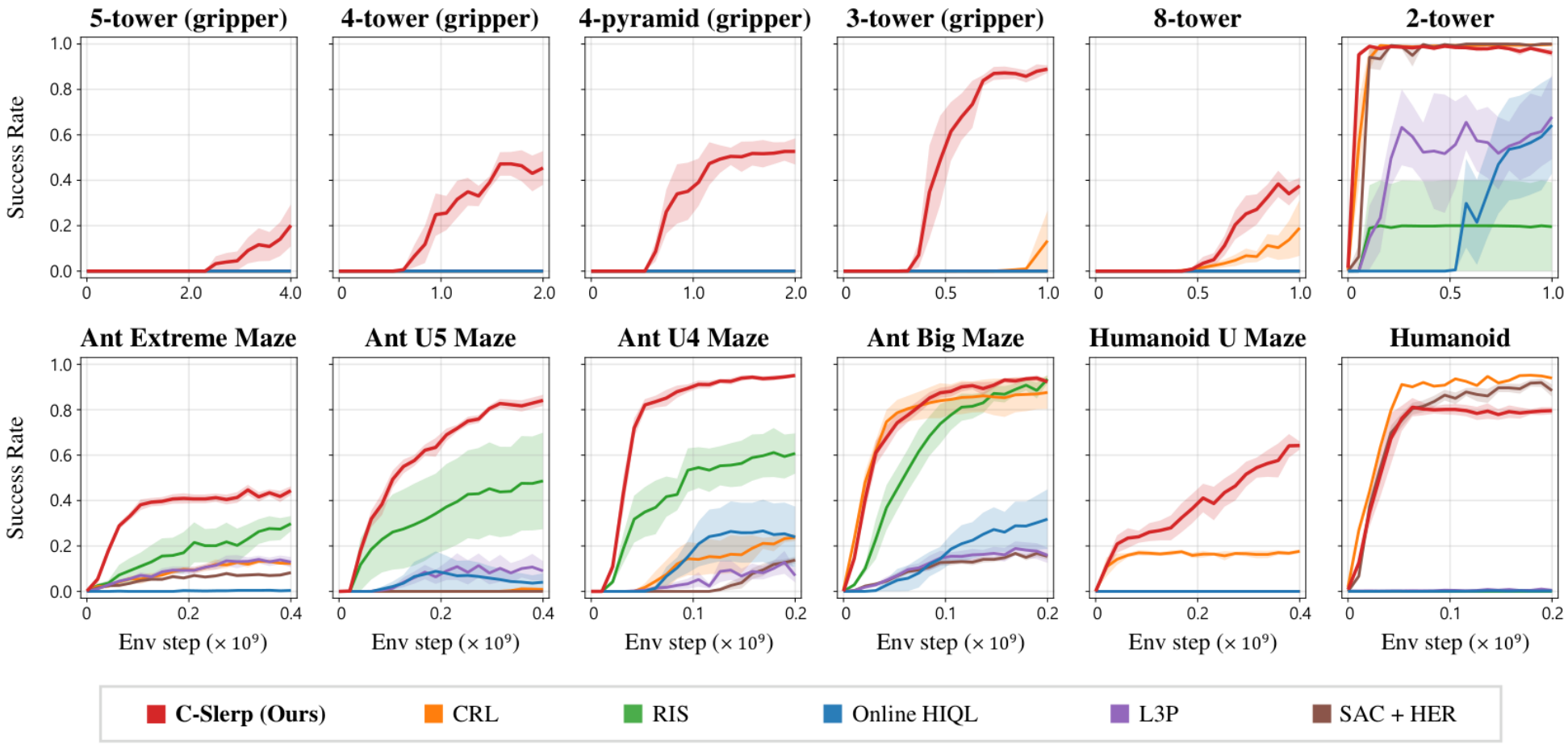
Experimental Results 👀

Benchmark Environments



Six manipulation tasks (**BuilderBench**) and six locomotion tasks (**JaxGCRL**)

Main Results : BuilderBench (top) and JaxGCRL (bottom)



Our method solves long-horizon tasks where baselines **never achieve a single success**.

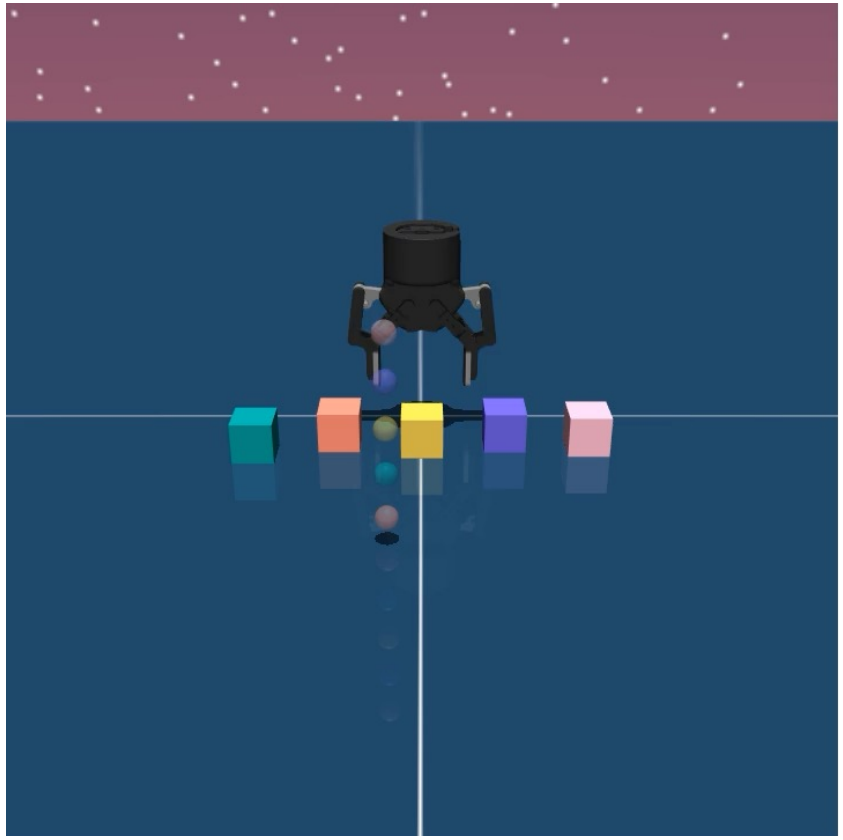
Latent Visualization : JAXGCRL

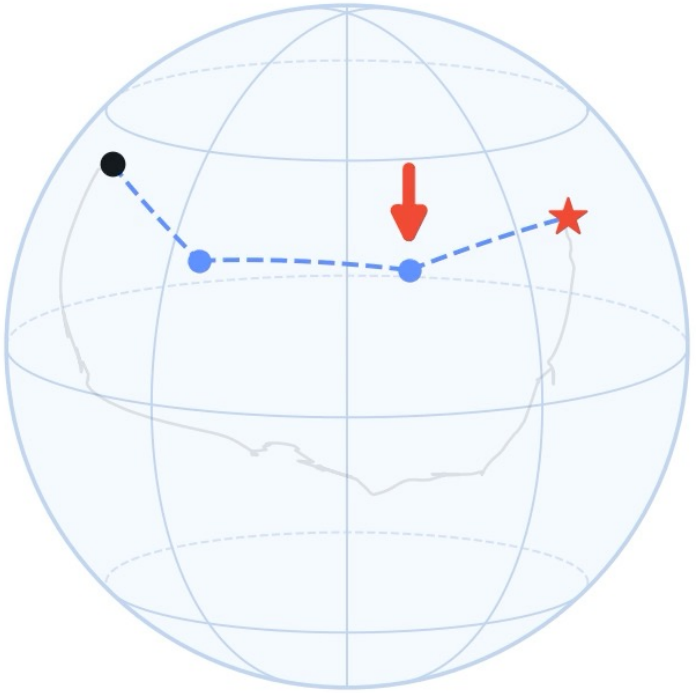




● Current state $\psi(s)$ ★ Goal $\psi(g)$ ● Inferred waypoints $z_{1:K}$ → Selected subgoal

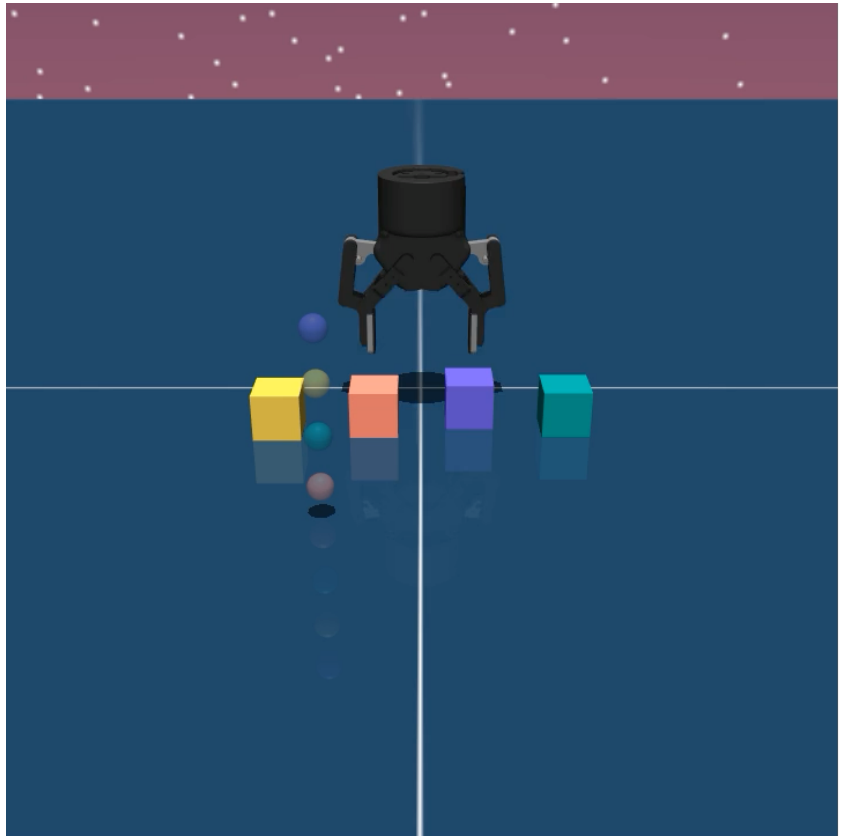
Latent Visualization : BuilderBench (5-tower)

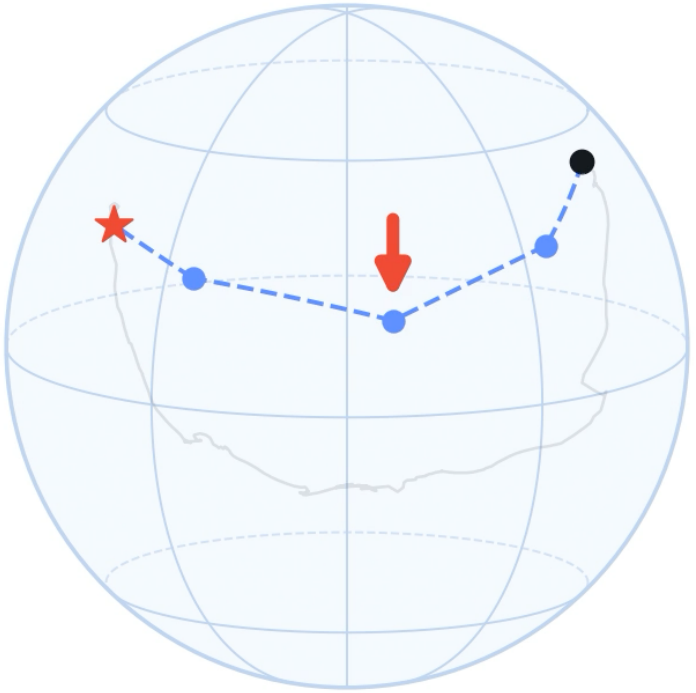




● Current state $\psi(s)$ ★ Goal $\psi(g)$ ● Inferred waypoints $z_{1:K}$ → Selected subgoal

Latent Visualization : BuilderBench (4-tower)

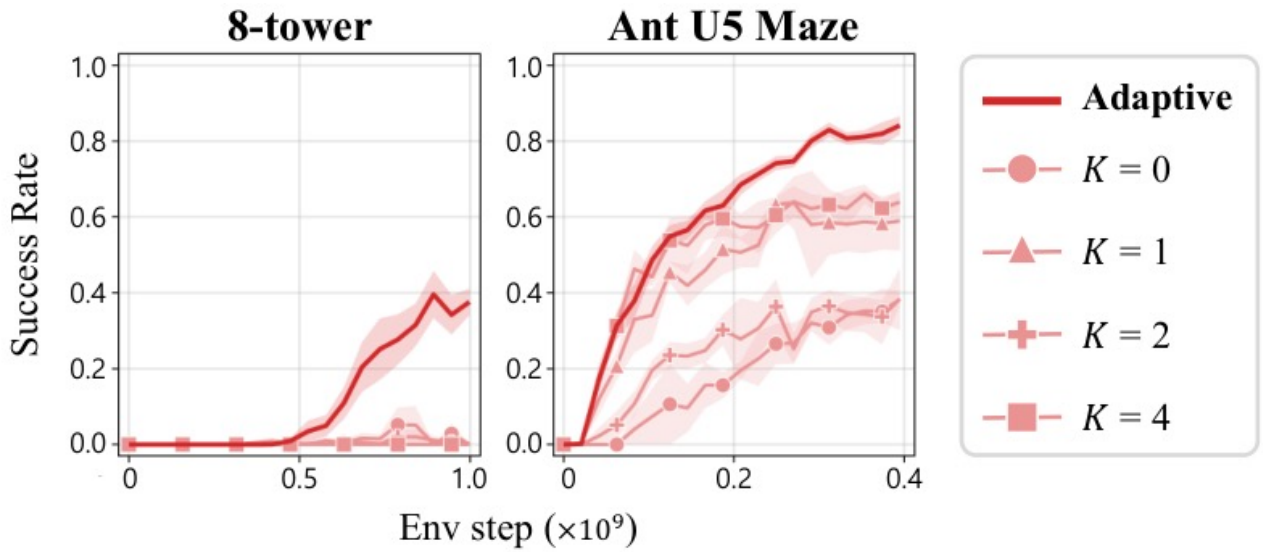




● Current state $\psi(s)$ ★ Goal $\psi(g)$ ● Inferred waypoints $z_{1:K}$ → Selected subgoal

Does adaptivity matter? : ablation on *K*-selection

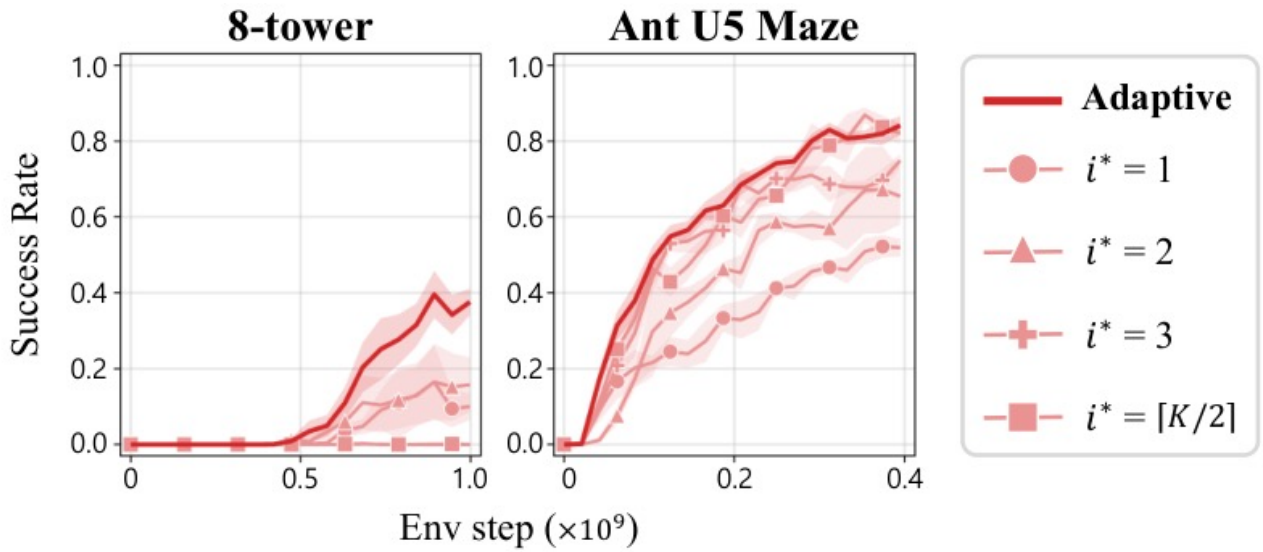
Fixing the number of waypoints — at *any* value — substantially **degrades** performance.



- More waypoints is *not* consistently better.
- The right *K* also changes *within* an episode as the agent nears the goal.

Does adaptivity matter? : ablation on *i*-selection

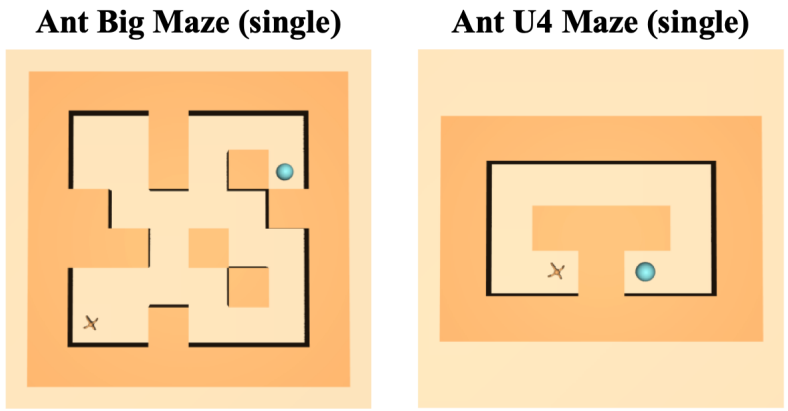
The best fixed choice of *i* **differs** across environments; **adaptive selection is best** in both.



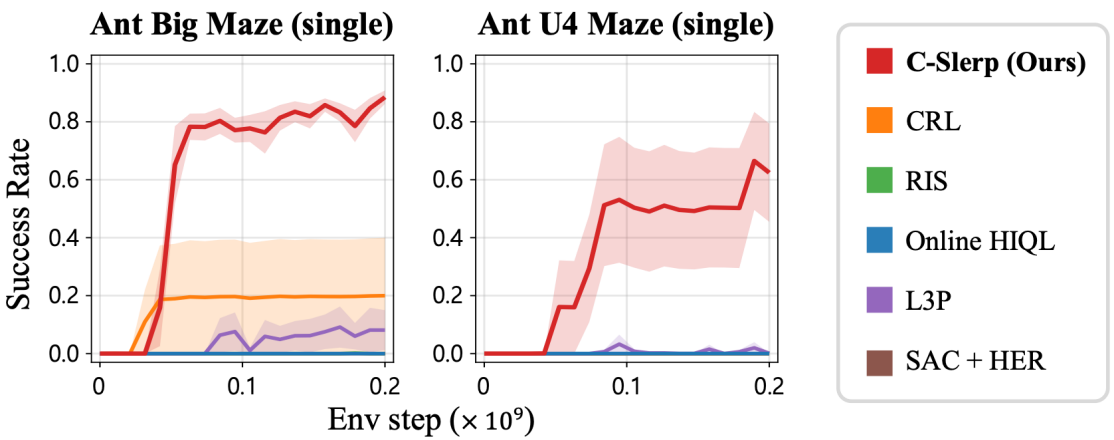
- Early waypoints are safe but slow; late ones make progress but may be unreachable.
- Adaptive *i*-selection method picks the furthest reachable one.

Exploration in single-goal settings

With only **one distant goal ever provided**, our method still achieves a high success rate — **directed exploration** without a goal curriculum.



(a) Single-goal environments



(b) Learning curves

Takeaways

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1 **Adaptive planning.** Adapting task decomposition to the goal-reaching problem is key to the benefits of hierarchy — fixing the number of planning stages substantially degrades performance.

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2 **Exploration.** Our method solves long-horizon tasks where all evaluated baselines fail to achieve a single success, without a goal curriculum or dense rewards.

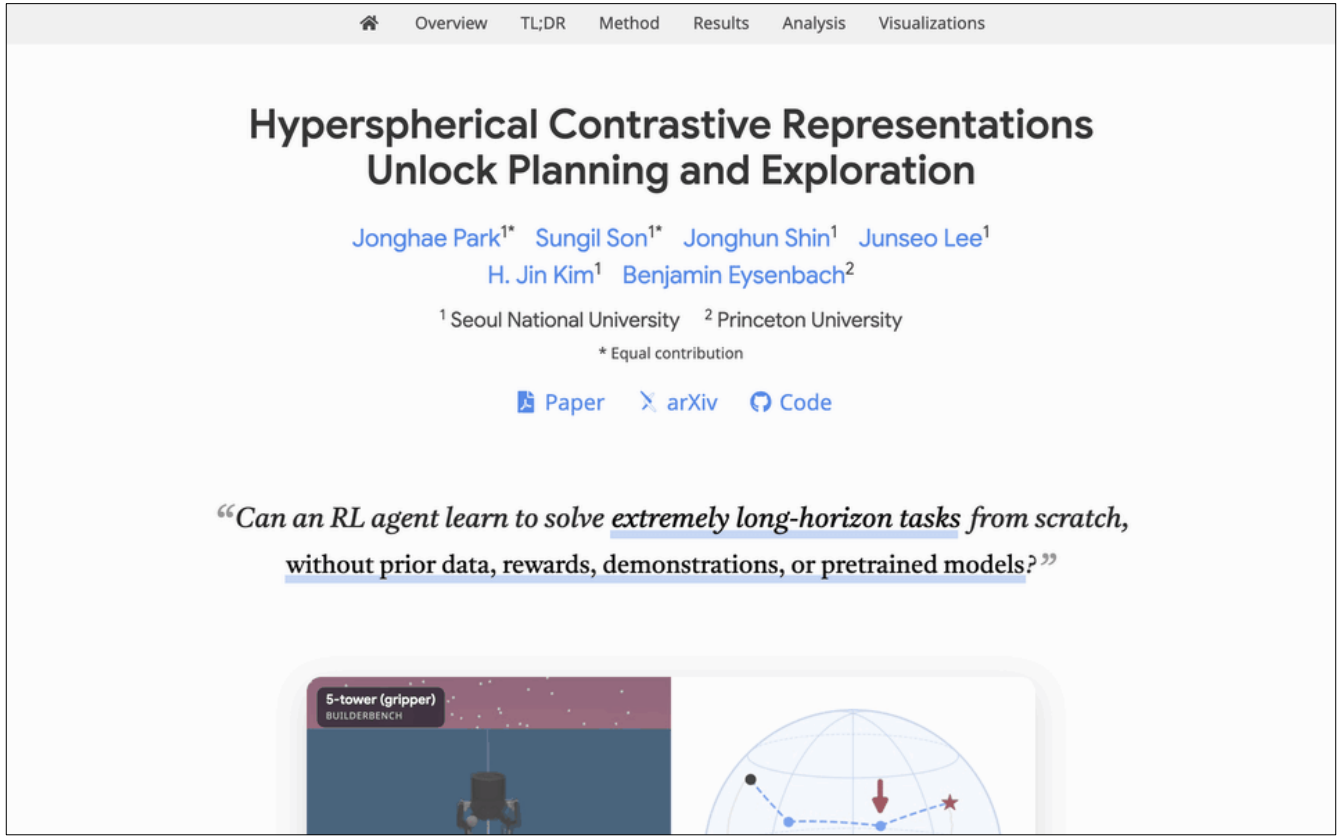
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1 **Adaptive planning.** Adapting task decomposition to the goal-reaching problem is key to the benefits of hierarchy — fixing the number of planning stages substantially degrades performance.

2 **Exploration.** Our method solves long-horizon tasks where all evaluated baselines fail to achieve a single success, without a goal curriculum or dense rewards.

3 **No extra architecture.** Planning follows directly from spherical linear interpolation in the learned representation space — no separate high-level network, no planning graph.

For more results & derivations ..



https://jonghaepark.github.io/c-slerp/

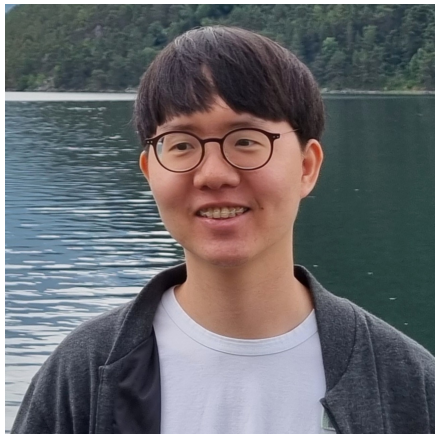
Thanks to my *collaborators*!



Sungil



Jonghun



Junseo



H. Jin Kim



Benjamin Eysenbach

Thank you

Any questions?